CEYLON RAIL : AI-DRIVEN INTELLIGENT RAILWAY OPERATIONS SYSTEM FOR SRI LANKA

INDIVIDUAL THESIS

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REAL TIME TRAIN TRACKING AND DELAY PREDICTION

R26-IT-127

Report submitted in partial fulfilment of the requirements for the

B.Sc. (Hons) Degree in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

Oct 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Railway passengers require timely and reliable information about train movements and delays to plan their journeys effectively. However, limitations in the availability, consistency, and timeliness of railway operational data can make accurate train tracking and delay prediction challenging. This research and development project aimed to develop a real-time train tracking and delay prediction component that processes available railway information, displays the latest available train movement status, and investigates machine learning techniques for predicting delays. The methodology involved identifying and preparing relevant railway data, validating train identifiers and timestamps, designing a data-processing workflow, implementing train-tracking functionality, and developing and evaluating machine learning models. An LSTM-based regression approach was implemented to explore delay prediction, while a classification model was evaluated using precision, recall, F1-score, accuracy, and a confusion matrix. The classification evaluation used 1,065 test observations. The model classified all observations as Moderate Delay, achieving an accuracy of approximately 34.5%, while failing to correctly identify the On-time / Low Delay and High Delay classes. This result highlighted an important limitation in the model's ability to distinguish between different delay categories and demonstrated the importance of class-sensitive evaluation. The project established a technical foundation for combining train-tracking information with predictive analysis in a railway information system. Its principal contribution is the implementation of a data-processing and tracking workflow, the investigation of machine learning for delay prediction, and the identification of areas requiring further improvement. Future work should focus on improving classification performance, incorporating more comprehensive and timely railway data, comparing alternative prediction methods, and evaluating the system under realistic operational conditions.

Keywords: Real-Time Train Tracking, Train Delay Prediction, Machine Learning, LSTM, Railway Information Systems

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my project supervisor for the guidance, advice, and constructive feedback provided throughout this research and development project. The support and recommendations received were valuable in shaping the direction of the work and improving the development and evaluation of the Real-Time Train Tracking and Delay Prediction System.

I also wish to thank the academic staff and the institution for providing the learning environment and resources necessary to undertake this project. My appreciation extends to everyone who helped, encouragement, or technical guidance during the development and documentation of this individual component.

Finally, I am grateful to my family and friends for their patience, encouragement, and support throughout the completion of this research and thesis.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
ETA	Estimated Time of Arrival
F1-score	Harmonic Mean of Precision and Recall
GIS	Geographic Information System
GPS	Global Positioning System
GSM	Global System for Mobile Communications
IoT	Internet of Things
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
MAPE	Mean Absolute Percentage Error
MLP	Multilayer Perceptron
RMSE	Root Mean Squared Error
RQ	Research Question

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1 INTRODUCTION

1.1 Overall Project Context

Railway transportation is an important part of Sri Lanka's public transport network, connecting urban centers, suburban communities, and regional destinations at affordable costs. Service reliability affects passengers' daily activities and journey planning. Train delays and limited real-time information can cause longer waiting times, missed connections, and uncertainty about arrival times.

The project proposal identifies limited real-time train visibility and unreliable delay prediction as key challenges. Static timetables and delayed announcements may not reflect actual railway conditions. Therefore, railway staff need timely and accurate information to monitor train movements, predict delays, and manage disruptions effectively.

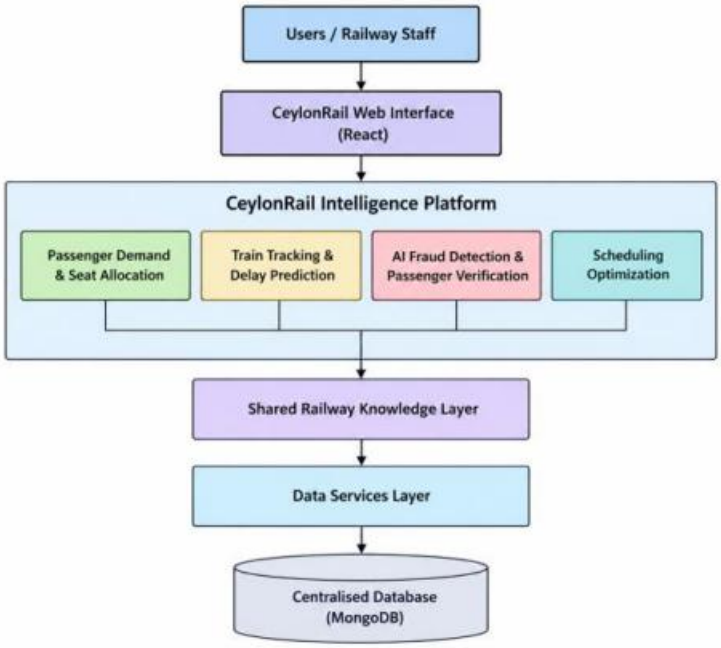


Figure 1 - System Overview

The wider AI-Driven Intelligent Railway Operations System comprises four components: passenger-demand forecasting and adaptive seat allocation, real-time train tracking and delay prediction, ticket-fraud detection and risk analysis, and train-scheduling optimization and capacity management. Together, these components aim to support data-driven railway management.

This thesis focuses exclusively on the Real-Time Train Tracking and Delay Prediction component, which uses train locations, timestamps, timetable information and relevant operational data to monitor progress and estimate delays or arrival times. Its data requirements, implementation and evaluation are considered separately from the other project components.

1.2 Individual component context

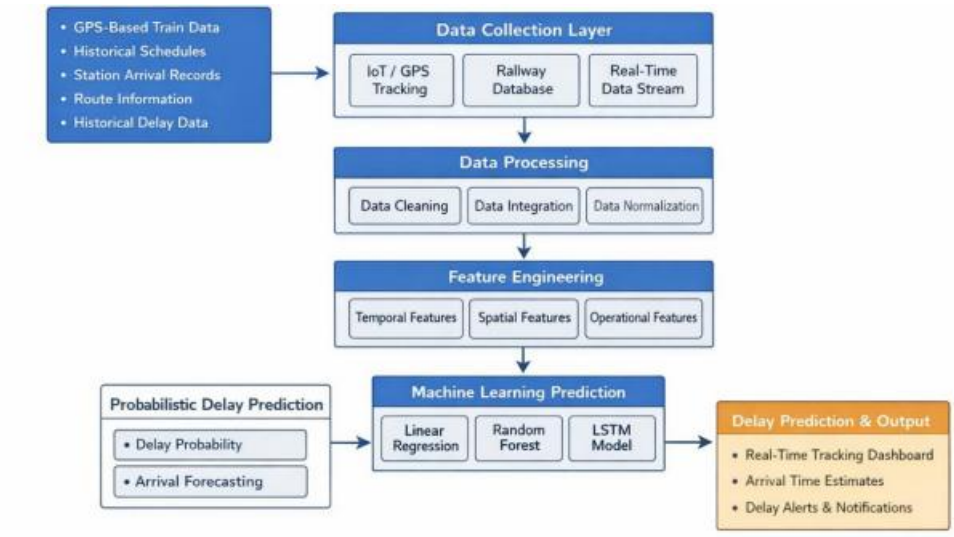


Figure 2 - component overview

The individual component combines real-time train tracking and delay prediction. Tracking identifies and displays a train's latest reported position and movement status using available GPS coordinates, timestamps, train identifiers, routes and station records. Location observations must be matched to the correct train and route, and information freshness should be indicated because outdated positions do not represent live locations.

Delay prediction estimates journey progress using historical and current information. Arrival delay is calculated as actual arrival time minus scheduled arrival time: positive values indicate late arrivals, zero indicates an on-time arrival, and negative values indicate early arrivals. The implementation must consistently define whether it predicts delay minutes, arrival times or delay categories.

Relevant data identified in the proposal include historical timetables, arrival and departure records, route information, station dwell times and delay records. Available real-time location data may provide additional context. These sources require cleaning, time-stamp alignment and validation of identifiers, duplicates, missing values and unusual records. Potential features include time of day, journey segment, elapsed

journey time, inter-station distance, previous delay and dwell time, subject to actual data availability.

The proposal identifies Linear Regression, Random Forest Regression and Long Short-Term Memory (LSTM) networks as candidate models. Linear Regression provides a baseline, Random Forest can capture nonlinear relationships, and LSTM can learn sequential patterns. Model selection must be based on consistent experiments and evaluation results, distinguishing proposed methods from those actually implemented and tested.

Probabilistic prediction is another proposed goal. Unlike point estimates, it communicates uncertainty through outputs such as the probability of exceeding a delay threshold or an interval of plausible arrival times. Point predictions alone do not constitute probabilistic prediction; therefore, uncertainty estimation must be implemented and evaluated before such claims are made.

1.3 Problem statement

Passengers and railway operators need reliable information about train progress and expected arrival times. When location updates are unavailable, infrequent or delayed, users must rely on timetables that do not reflect current conditions. Although announcements provide updates, they may not offer continuous route-level monitoring or consistent arrival predictions.

The technical problem is to process available train movement and schedule information to display current progress and generate useful delay estimates. This involves validating and matching location observations to trains and routes, presenting their reported status, and transforming historical and current data into suitable prediction-model inputs.

Prediction is challenging because delays may result from variable station dwell times, congestion, signaling or technical problems, weather and other operational disruptions. These factors may interact, while datasets may not record every cause. If live feeds or detailed historical records are unavailable, prepared datasets or simulated location updates may be required. Such prototypes demonstrate the processing workflow but cannot establish performance under fully live railway conditions.

Therefore, this research addresses the need for an integrated, evaluable component that uses available railway location, timetable and operational data to display train progress and estimate delays or arrival times, while explicitly communicating limitations in data freshness, accuracy and uncertainty. Evaluation must include appropriate quantitative metrics and software tests rather than interface demonstrations alone.

1.4 Research gap

The project proposal reviews related work on IoT-based monitoring, machine-learning-supported railway management, GPS-based tracking and arrival-time prediction, infrastructure inspection and predictive transport analytics. It suggests that monitoring, prediction and infrastructure management are frequently addressed separately. References [1]–[9] provide the proposal's initial research context and must be verified against the original publications before submission.

The first identified gap is the limited integration of current train movement information with delay prediction. Tracking systems may display locations without estimating arrival times, while historical-data-based models may not incorporate recent train progress. Integrating these functions could provide more consistent location and prediction updates, provided the data are sufficiently accurate.

The second gap concerns uncertainty. A single arrival-time estimate may imply unjustified certainty when journey times and operating conditions vary. Probabilistic methods could represent uncertainty arising from travel times, dwell times and congestion. However, their reliability must be evaluated; if the implementation produces only point estimates or delay categories, probabilistic prediction must remain a limitation or future extension.

Data availability is another context-specific challenge. In Sri Lanka, the feasibility of the proposed approach depends on documenting dataset sources, coverage, completeness and whether location updates are live, historical or simulated. This enables an accurate assessment of the prototype's capabilities and potential operational applicability.

Accordingly, this thesis addresses an implementation and evaluation gap by developing and accessing a workflow that connects train tracking with delay prediction using available data. Model performance will be established experimentally rather than assumed in advance.

1.5 Aim and objectives

The aim is to design, implement and evaluate a real-time train tracking and delay prediction component that processes available railway movement and timetable information to display train progress and estimate delays or arrival times, while documenting its accuracy and limitations.

The specific objectives are:

1. Identify and prepare available timetable, timestamp, route, station and delay data required for tracking and prediction.
2. Develop a data-processing workflow that validates and aligns train identifiers, location observations, timestamps and operational features.

3. Implement train tracking that displays the latest available position or movement status and indicates data source and freshness where possible.
4. Develop and evaluate suitable prediction models, including Linear Regression, Random Forest Regression and/or LSTM where actually implemented.
5. Evaluate performance using MAE and RMSE for regression, or precision, recall, F1-score, confusion matrices and accuracy for classification.
6. Test the tracking and prediction workflow, document integration, and identify limitations in data availability, accuracy, latency and deployment.
7. Recommend improvements to reliability, uncertainty communication and future operational suitability.

1.6 Research questions or hypotheses

The study addresses the following research questions:

- **RQ1:** How can available train location, timestamp, timetable and route data be processed to present a coherent view of train progress?
- **RQ2:** How accurately can implemented models estimate train delays or arrival times using the available dataset?
- **RQ3:** Which evaluated model provides the most suitable results under the selected testing procedure, and what evidence supports the comparison?
- **RQ4:** What limitations arise from data quality, missing observations, dataset coverage and differences between offline evaluation and live operation?
- **RQ5:** To what extent does the component integrate tracking and delay prediction to support passenger information or railway operational awareness?

If the experimental design supports formal hypothesis testing, H1 states that a selected machine-learning model produces lower test-set prediction error than a defined baseline. H0 states that it does not. The baseline, data split, error metric and comparison procedure must be specified consistently. For prototype-focused research, the research questions may be more appropriate than inferential hypotheses.

1.7 Scope limitations and assumptions

This thesis covers only the Real-Time Tracking and Delay Prediction component of the wider AI-Driven Intelligent Railway Operations System. Its scope includes data preparation, train-progress tracking, delay or arrival-time prediction, output presentation and technical evaluation. Passenger-demand forecasting, adaptive seat allocation, ticket-fraud detection and scheduling optimization remain outside its core scope.

Inputs may include historical timetables, actual or recorded arrival and departure times, station and route information, delay records, GPS coordinates or simulated updates. The methodology must identify the exact dataset, source, observation period, usable record count, target variable and preprocessing decisions. Live APIs, GPS devices or official operational databases must not be claimed unless used.

Prediction evaluation depends on the implemented target. Delay-minute predictions, scheduled-versus-actual arrival differences and categorical delay predictions require clearly defined outputs. Classification thresholds and class distributions must be documented where applicable. Regression and classification metrics measure different outcomes and must not be treated as interchangeable.

Key assumptions include consistent timestamp formats and time zones, correct train identifiers, and sufficiently accurate recorded arrival and departure times. Assumptions about missing data, route matching and simulated updates must be documented. Historical or simulated datasets may not represent all real-world disruptions, and their results cannot establish equivalent live performance.

Prediction quality depends on data quantity, coverage and reliability, while unrecorded delay causes may limit model performance. A prototype interface alone cannot demonstrate operational improvement, passenger satisfaction or reduced railway delays; these require separate user studies or deployment evidence. Security, privacy and responsible handling of operational information are also important for future live-data integration.

1.8 Individual contribution

The individual contribution is the design, implementation and evaluation of the Real-Time Train Tracking and Delay Prediction component. It focuses on preparing railway data, engineering relevant features, implementing tracking and prediction functions, evaluating outputs and documenting limitations.

Evidence of individual work may include dataset-preparation pipelines, feature definitions, training notebooks or scripts, saved models, evaluation tables, confusion matrices, regression-error results, tracking visualizations, interface screenshots, test cases and integration documentation. The thesis must distinguish personally developed artefacts from teammates' contributions and reused tools.

The proposal identifies Linear Regression, Random Forest Regression and LSTM as candidate models. Only models actually trained and evaluated may be reported as completed work. Similarly, GPS monitoring and IoT-enabled acquisition must be distinguished from recorded-data playback or simulated tracking, depending on what was implemented.

The technical contribution is an integrated prototype connecting train-progress information with delay-related outputs. The research contribution is the evaluation

evidence showing what worked, prediction strengths and weaknesses, metric-based performance, and required improvements. Negative results remain valid when transparently reported. Claims of operational improvement, superior accuracy or probabilistic reliability require supporting experimental evidence.

1.9 Thesis structure

This thesis contains nine chapters. Chapter 1 introduces the wider project, individual component, problem statement, research gap, aim, objectives, research questions, scope and contribution.

Chapter 2 reviews railway tracking, arrival-time estimation, delay prediction, machine learning and uncertainty-aware forecasting, establishing the conceptual foundation and verifying relevant references.

Chapter 3 explains the methodology, including research design, data sources, preprocessing, feature engineering, model development, evaluation metrics, analysis, ethical considerations and reproducibility. Chapter 4 presents requirements, system architecture, design alternatives, integration and data flow.

Chapter 5 describes the implementation environment, technologies, tracking and prediction functions, integration, testing and deployment. Chapter 6 presents actual model metrics, tracking tests, visualizations and any completed stakeholder or integration validation. Chapter 7 interprets the findings against literature and baselines, explains individual contributions, and discusses limitations and threats to validity.

Chapter 8 examines potential users, value, competitive position, sustainability, scalability, intellectual property and relevant Sustainable Development Goals, distinguishing potential benefits from demonstrated outcomes. Chapter 9 summarizes findings, objective achievement, individual contributions and future directions. References and appendices provide cited sources, extended results, test evidence and required academic-integrity or AI-use disclosures.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter reviews research and technologies related to real-time train tracking and delay prediction, establishing theoretical foundation and identifying challenges in monitoring train movements and predicting arrival times. Particular attention is given to Sri Lankan railway research and international studies on machine learning, historical travel-time analysis, online prediction and probabilistic delay estimation.

The review covers fifteen publications from 2010–2025 addressing railway tracking, arrival-time estimation, delay prediction, railway supervision and intelligent management. These include GPS/GSM-based systems, passenger information services, machine-learning models and uncertainty-aware prediction approaches.

The chapter examines theoretical foundations, related studies, existing systems and techniques, and the research gap and conceptual framework. It addresses five questions:

- How do railway systems monitor train locations and communicate operational information?
- What techniques predict train arrival times and delays?
- How can machine learning support delay prediction?
- How can prediction uncertainty be represented?
- What opportunities exist for integrating tracking and prediction in Sri Lanka?

This is a focused narrative review, not a systematic review of all available publications. The discussion uses the supplied references and available publication information; findings not verified through full-text examination are not assumed.

2.2 Domain and theoretical foundations

2.2.1 Real-time train tracking

Real-time train tracking obtains and presents a train's current or most recently reported position and movement status using information such as GPS coordinates, timestamps, train identifiers, speed, routes and station locations.

Typically, location observations are collected onboard or through another positioning source, transmitted to a central server and displayed on a map or dashboard. GPS and GSM technologies have been investigated for railway tracking [1], [2].

Tracking accuracy depends on positioning quality, communication availability, update frequency and data reliability. An effective system must associate observations with the correct train and route, display the latest position and timestamp, and handle missing or invalid updates. Showing timestamps is essential to distinguish current positions from outdated information.

2.2.2 Train arrival-time and delay prediction

Arrival-time prediction estimates when a train will reach a future station, while delay prediction measures the difference between an observed or predicted event time and its scheduled time.

Prediction may use historical journey durations, timetables, recent location observations, station dwell times and other operational variables. Historical records reveal recurring patterns, while recent observations can update estimates during a journey. Reliability depends on data quality and modelling assumptions.

TrainKo, developed for Sri Lanka, illustrates the use of train movement information to predict arrival times at upcoming stations and notify passengers [3].

2.2.3 Machine learning for railway prediction

Machine learning identifies relationships between input features and target variables using historical examples. Railway prediction features may include scheduled times, previous delays, route segments, station dwell times and recent movement observations.

Linear Regression: Models linear relationships and provide a relatively interpretable baseline.

Random Forest Regression: Combines decision trees to capture nonlinear relationships, depending on data quality and representativeness.

Long Short-Term Memory (LSTM): Uses recurrent neural network architecture to learn sequential dependencies, requiring suitable temporal data and preparation.

Gradient Boosting and Multilayer Perceptron models have also been investigated for railway delay prediction [11], [12]. Model selection should therefore depend on

experimental results rather than assumptions that more complex algorithms perform better.

2.2.4 Evaluation of prediction models

Models must be evaluated on data not used during training to determine their predictive performance. Two common regression metrics are Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

For delay classification into categories such as low, moderate and high, precision, recall, F1-score and confusion matrices are appropriate. Precision measures the proportion of predicted class instances that are correct; recall measures the proportion of actual class instances correctly identified; and F1-score combines both.

Regression and classification represent different prediction tasks and must be evaluated separately using appropriate metrics.

2.2.5 Probabilistic delay prediction

Point prediction provides a single estimated arrival time or delay, whereas probabilistic prediction represents uncertainty through outputs such as prediction intervals or the probability of exceeding a delay threshold.

This distinction matters because journey durations and station dwell times vary. Spanninger, Büchel and Corman investigated probabilistic modelling of train-delay evolution using stochastic processes and historical movement observations [14].

A model must generate and evaluate an appropriate uncertainty representation before it can be described as probabilistic.

2.3 Related research

2.3.1 GPS/GSM-based train tracking systems

Jayakody and colleagues investigated GPS/GSM-based train tracking using mobile communication networks [1]. Their work illustrates how GPS can provide geographic observations and GSM can transmit them to a central service for monitoring. Practical effectiveness depends on positioning accuracy and communication coverage.

Sayooran and colleagues examined an outstation railway supervision system for Sri Lanka Railways [2]. This research extends the monitoring context beyond passenger-facing location displays, although the supplied bibliographic information is

insufficient to compare its implementation or measured performance with the present component.

Together, these studies demonstrate the relevance of information technology to railway monitoring and the distinction between collecting operational data and using it in applications.

2.3.2 Train arrival-time prediction in Sri Lanka

Jayawardena, Sachith and Wijekoon introduced TrainKo, a Sri Lankan arrival-time prediction system designed to estimate arrivals at upcoming stations and notify subscribers [3]. Its reported pilot involved two adjacent stations, using journey speed observations to estimate arrival times. Although this demonstrates a practical evaluation approach, a limited route and number of journeys cannot establish network-wide performance.

Jayasinghe and Kathriarachchi proposed a Sri Lankan real-time tracking system using GPS, GSM, an Arduino microcontroller and network services [4]. Its central application supports map-based monitoring and operational decisions. However, displaying current position differs from predicting future arrival times; integration requires connecting tracking observations with a prediction model and evaluating both functions.

Silva and colleagues presented SmartRail, an optimized train-management system for Sri Lanka Railways [5]. This work provides a broader operational perspective, although specific performance claims require verification against the full publication.

Wanniarachchi and colleagues investigated a *Real-Time Train Tracking and Time Prediction System for Railway System in Sri Lanka* [6]. Its subject closely matches the general problem addressed here, demonstrating that combining tracking and prediction has already been explored locally. The present research must therefore establish its contribution through demonstrable differences in implementation, datasets, model selection, integration or evaluation rather than claiming the combination is entirely new.

2.3.3 Historical travel times and online prediction

Jayawardena, Sachith and Wijekoon introduced TrainKo, a Sri Lankan arrival-time prediction system designed to estimate arrivals at upcoming stations and notify subscribers [3]. Its reported pilot involved two adjacent stations, using journey speed observations to estimate arrival times. Although this demonstrates a practical

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2.3.3 Historical Travel Times and Online Prediction

Pongnumkul and colleagues investigated improving Thai passenger-train arrival predictions using historical travel times [7]. Historical journey durations can reveal patterns beyond published timetables, although their usefulness depends on similarities between past and current routes, operating conditions and disruptions.

Kecman and Goverde studied adaptive, data-driven online prediction of train event times [8]. Their work supports updating forecasts as new operational observations arrive rather than relying exclusively on initial estimates. Update frequency, timestamp freshness and missing observations are therefore important design considerations.

Oneto and colleagues examined railway delay prediction at network scale using big-data analytics [9]. Their research highlights the additional processing, data management and validation needed to extend predictions from individual trains or routes to larger networks.

These studies establish the importance of historical information, recent observations and scalability, but their findings require local validation before being applied to Sri Lankan railway operations.

2.3.4 Machine-learning approaches to train delay prediction

Hauck and Kliewer examined machine learning and data analytics for railway delay prediction [10]. Such models learn relationships between available features and observed delays, but their performance depends on target definitions, data preparation, feature availability and evaluation design.

Lapamonpinyo, Derrible and Corman investigated real-time passenger-train delay prediction on Amtrak routes using Random Forest, Gradient Boosting Machine and Multilayer Perceptron models [11]. Their work supports comparative evaluation rather than selecting models solely on complexity. However, differences in railway networks, timetables and operating conditions mean that Amtrak results cannot automatically be generalised to Sri Lanka.

Vafaei and Yaghini investigated online prediction of station arrival and departure times on the Tehran–Mashhad route using timetable and operational data from 2018–2020 [12]. They evaluated Artificial Neural Networks, Random Forest Regression, Gradient Boosting Regression and Extreme Gradient Boosting Regression, selecting an approach based on experimental results.

This research highlights the importance of specifying routes, observation periods and prediction targets. For the present thesis, Linear Regression, Random Forest Regression and LSTM remain candidates identified in the project proposal; only models actually implemented and tested should appear in the final comparison.

2.3.5 Alternative and probabilistic prediction techniques

Padma and colleagues proposed a robust real-time train-delay prediction system using fuzzy-based learning [13]. Fuzzy methods can represent approximate relationships and imprecise categories, but their suitability depends on implementation and evaluation. The reference supplied does not establish that this approach outperforms the models considered in this thesis.

Spanninger, Büchel and Corman investigated probabilistic train-delay evolution using probability distributions for arrival and departure delays [14]. Their work is relevant to uncertainty-aware prediction, which could communicate the likelihood of different arrival outcomes when travel times and station dwell times vary. Unlike deterministic models that estimate single values, probabilistic models characterise possible outcomes and their likelihood.

Meesit and Suwannasri investigated real-time delay prediction for single-track intercity railways [15]. Shared track sections can create interactions between train movements and influence delay development. However, the available bibliographic information does not establish the specific algorithm or numerical results, so this study is treated as a relevant research direction rather than evidence of a particular accuracy level.

2.4 Existing systems, products and techniques

Existing approaches differ in their data requirements, architectures, prediction methods and intended users.

2.4.1 GPS/GSM-based tracking architecture

GPS/GSM systems collect train locations and transmit them to central applications. The Sri Lankan architecture investigated by Jayasinghe and Kathriarachchi uses GPS, GSM and a microcontroller for centralized monitoring [4].

Its advantages include centralized location processing and map-based presentation, but implementation requires positioning equipment, communication channels and a service to process updates. The present component must reflect its actual data source: live feeds should be distinguished from recorded or simulated coordinates.

2.4.2 Passenger information and arrival-time estimation

Passenger-facing systems convert operational data into useful travel information. TrainKo [3] illustrates arrival prediction and passenger notifications. Effective interfaces should display estimated arrival times, delay status where available, and the last update timestamp.

Predictions remain estimates rather than guarantees, particularly when train progress changes after an estimate has been generated.

2.4.3 Machine-learning prediction techniques

Research has examined Random Forest, gradient boosting and neural-network methods for railway prediction [10]–[12]. Selection depends on the target variable, data availability, computational requirements and experimental performance.

Table 1 - Machine-learning prediction techniques

Technique	Main characteristic	Consideration for this research
Linear Regression	Models linear relationships	Useful as a baseline
Random Forest Regression	Combines decision trees	Can model non-linear feature relationships
LSTM	Learns patterns in sequential data	Requires suitable sequence preparation
Gradient Boosting	Builds an ensemble of predictive models	A possible comparison method if implemented
Probabilistic prediction	Represents uncertainty in possible outcomes	Requires explicit uncertainty estimation and validation

These techniques produce different outputs. Regression estimates numerical values, classification assigns predefined categories, and probabilistic methods quantify uncertainty. Fair comparisons require consistent target definitions and appropriate test data. MAE and RMSE suit regression, while precision, recall, F1-score and confusion matrices suit classification.

2.4.4 Online and adaptive prediction

Online prediction updates forecasts when new operational information becomes available. Kecman and Goverde [8] provide a relevant foundation for adaptive train-event prediction.

In this component, updated locations could refresh arrival estimates, provided timestamps are reliable and missing or outdated observations are handled appropriately. Recording each prediction's generation time and input information supports evaluation of its freshness and relevance.

2.4.5 Probabilistic methods and uncertainty

Probabilistic modelling offers an alternative to single-value predictions. Spanninger and colleagues [14] demonstrate its relevance to train-delay evolution.

Future extensions could provide prediction intervals or probabilities of exceeding defined delay thresholds, helping users understand uncertainty. However, probabilistic functionality must not be claimed unless uncertainty estimates are generated and evaluated. Single-value or categorical outputs must be described accurately.

2.4.6 Comparison of existing approaches

The literature establishes that tracking and prediction address complementary needs: tracking reports a train's position, while prediction estimates its future progress. Historical travel-time analysis, machine learning, online adaptation and probabilistic modelling provide alternative approaches with different data requirements and outputs.

The present research therefore focuses on processing available railway information, displaying train progress and evaluating delay predictions. Its effectiveness must be demonstrated through implementation and experimental results rather than inferred from related systems' stated objectives.

2.5 Gap synthesis and conceptual framework

2.5.1 Synthesis of the research gap

The reviewed publications demonstrate that tracking and delay prediction have been investigated in Sri Lanka and internationally. Early studies addressed GPS/GSM tracking [1], railway supervision [2] and passenger arrival prediction [3]. Later Sri Lankan work examined tracking architectures [4], railway management [5] and combined tracking and time prediction [6].

International research explored historical travel times [7], online event prediction [8], large-scale railway analytics [9], machine-learning methods [10], [11], station-level prediction [12], fuzzy-based learning [13], probabilistic delay evolution [14] and single-track railway prediction [15].

Therefore, the gap cannot be defined as the complete absence of tracking or prediction systems. Instead, this thesis addresses the design, implementation and evaluation of a specific integrated component using the data available to the project.

The research focuses on six concerns:

1. **Tracking–prediction integration:** Connect the latest available train-progress information with delay or arrival-time estimates.
2. **Data preparation:** Consistently process timetables, timestamps, routes, stations and delay records.
3. **Model evaluation:** Evaluate implemented models using task-appropriate metrics.
4. **Operational transparency:** Distinguish live, recorded and simulated location data and document their limitations.
5. **Prediction limitations:** Analyze errors and determine the usefulness of predictions for the defined task.

6. **Future uncertainty estimation:** Investigate probabilistic prediction if uncertainty estimates are not yet implemented and validated.

The contribution is the evidence produced by developing and evaluating the selected component, not an unsupported claim that the underlying research problem has never been investigated.

2.5.2 Conceptual framework

The framework connects available railway data with tracking, prediction, output presentation and performance evaluation.

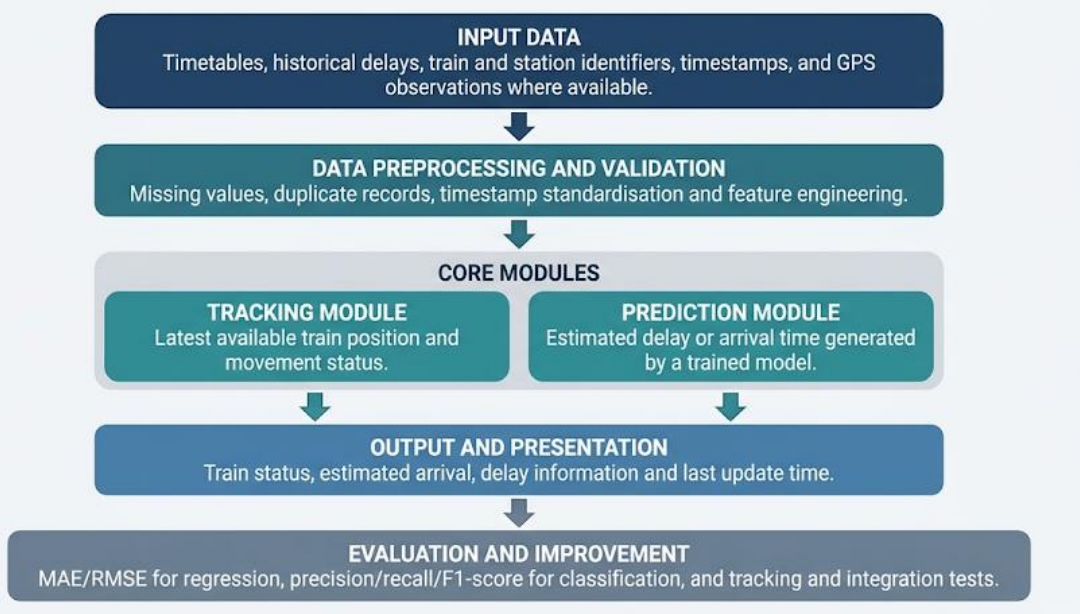


Figure 3 - Conceptual framework

The workflow begins by collecting or loading available railway data, validating identifiers and timestamps, and preparing relevant features. The tracking module processes location observations, while the prediction module estimates delay or arrival time. Results are presented through a dashboard or another interface, with predictions potentially refreshed as new observations arrive. Evaluation against observed data identifies areas for improvement.

The framework assumes that the data can reliably identify trains and prediction targets. Historical, recorded and live observations must be distinguished. Simulated positions support functional testing but cannot establish performance under actual railway operating conditions.

2.5.3 Relationship between the framework and the research objectives

The framework supports data preparation through preprocessing, train-progress presentation through tracking, and delay estimation through prediction. Its evaluation stage provides evidence of model accuracy and technical performance.

Prediction errors can be investigated in relation to data quality, target definitions, feature selection, training procedures and evaluation design. However, causes of poor performance must be established through diagnostic evidence rather than assumed.

In conclusion, the literature establishes the value of combining train tracking with data-driven delay prediction. Reliability depends on data quality, appropriate modelling, transparent outputs and rigorous evaluation. These findings provide the foundation for the research methodology and solution design presented in the following chapters.

3: RESEARCH METHODOLOGY

This chapter describes the methodology for developing and evaluating the proposed Real-Time Train Tracking and Delay Prediction System for the Sri Lankan Railway System. It covers research design, study materials, procedure, evaluation metrics, data analysis, ethical and security considerations, and reproducibility.

The system has two connected functions: tracking train movement using available location and operational data and predicting delays to provide timely information. Final implementation details must reflect the dataset, software, tracking hardware, and prediction models used. Where details are unavailable, the methodology below is a proposed approach and must not imply that data collection or experiments have already occurred.

3.1 Research design

This study adopts applied quantitative, system-development and experimental evaluation design. It develops a practical solution and evaluates numerical data such as train movement, scheduled and actual arrival times, delay duration, location updates, and prediction errors.

The system has two components:

1**. Real-time train tracking**: obtains available train-location or movement information, potentially including GPS coordinates, timestamps, station information, train identifiers, or operational records. It processes these inputs and presents the estimated or reported position through a map or status dashboard.

2. **Train delay prediction:** uses available historical and, where supported, current operational information to estimate future arrival time or delay. Statistical and machine-learning techniques may be considered, with selection based on data availability, performance, computational requirements, and suitability for railway operations.

Development and evaluation proceed iteratively:

- Identify requirements and available data.
- Prepare and validate data.
- Design and implement tracking and prediction components.
- Test components separately and together.
- Evaluate prediction performance and tracking behavior against defined criteria.
- Identify limitations and improvements.

Tracking and prediction must be evaluated separately: displaying the latest available position does not guarantee accurate arrival predictions. The study assesses feasibility and performance within the available data and experimental environment. If live feeds or physical tracking devices are unavailable, tracking must be evaluated using the data available, and the limitation stated. Historical or simulated data must not be described as verified live tracking.

3.2 Data participants or study materials

The study primarily uses train movement and operational data. Passengers or railway employees are not research participants when evaluation uses datasets and software alone. If interviews, surveys, or usability tests are conducted, participants and procedures must be described separately.

3.2.1 Data sources

The study may use the following data fields, depending on availability:

Table 2 - Data fields

Data category	Potential fields	Purpose
Train identification	Train ID, service or route identifier	Distinguish trains and associate records
Schedule information	Scheduled departure and arrival times, station sequence	Establish the expected timetable
Actual movement information	Actual departure and arrival times, recorded events	Calculate observed delays
Location information	Latitude, longitude, timestamp, reported station	Display or reconstruct train movement
Operational information	Route segment, direction, intermediate stops, where available	Represent train progress and operating context
Historical records	Previous journeys, observed travel times and delays	Train and evaluate prediction models
Contextual information	Weather, day of week, time of day, or other relevant factors, if available	Investigate factors associated with delays

These are candidate fields, not a claim that all are present. The thesis must identify each field's actual source, format, time coverage, sampling frequency, and limitations.

3.2.2 Study population and unit of analysis

The population comprises train journeys, station events, or location records in the available data. The unit depends on the task: journey-level analysis treats each journey as an observation; station-event analysis treats each arrival/departure as an observation; and location-update analysis treats each timestamped position as an observation. Station-event or journey-level records are generally more suitable for arrival-time prediction than isolated locations. Records from one journey must be linked consistently and kept out of both training and test sets to prevent leakage.

3.2.3 Inclusion and exclusion criteria

Include records containing the fields required for analysis and that can be interpreted reliably. Exclude or flag records with missing timestamps, invalid coordinates, inconsistent train identifiers, impossible time sequences, duplicate events, or unreliable targets. Document exclusions and any imputation or correction. Missing values must not automatically mean zero delay, and absent location updates do not prove that a train has stopped.

3.2.4 Data adequacy and limitations

Report the number of records and journeys, date range, routes and stations covered, missing-data proportion, and availability of actual arrival times. If the data cover only part of the Sri Lankan railway network, do not generalize findings to the entire network without evidence.

3.3 Experimental or study procedure

The procedure uses sequential stages to support controlled and traceable development and evaluation.

Stage 1: Requirements and system definition

Define intended users, inputs, outputs, and constraints. Tracking requirements may include the latest known position, train and route identification, and update timestamp. Prediction requirements may include estimated arrival time, expected delay, and communication of uncertainty. Separate functional requirements (what the system does) from performance requirements (for example, acceptable update age).

Stage 2: Data acquisition

Collect available historical timetable and operational records. Where supported, obtain live location updates from a configured GPS device or authorized feed. Retain timestamps and identifiers linking records to trains, journeys, or stations. Record data

provenance (live, historical, manually entered, or simulated) and standardise time zones and timestamp formats.

Stage 3: Data preprocessing

Check data types, missing values, duplicates, inconsistent identifiers, and implausible time/location values. Standardize timestamps, associate records with journeys, and validate station sequences where possible. Arrival delay is calculated as:

Arrival delay = Actual arrival time − Scheduled arrival time

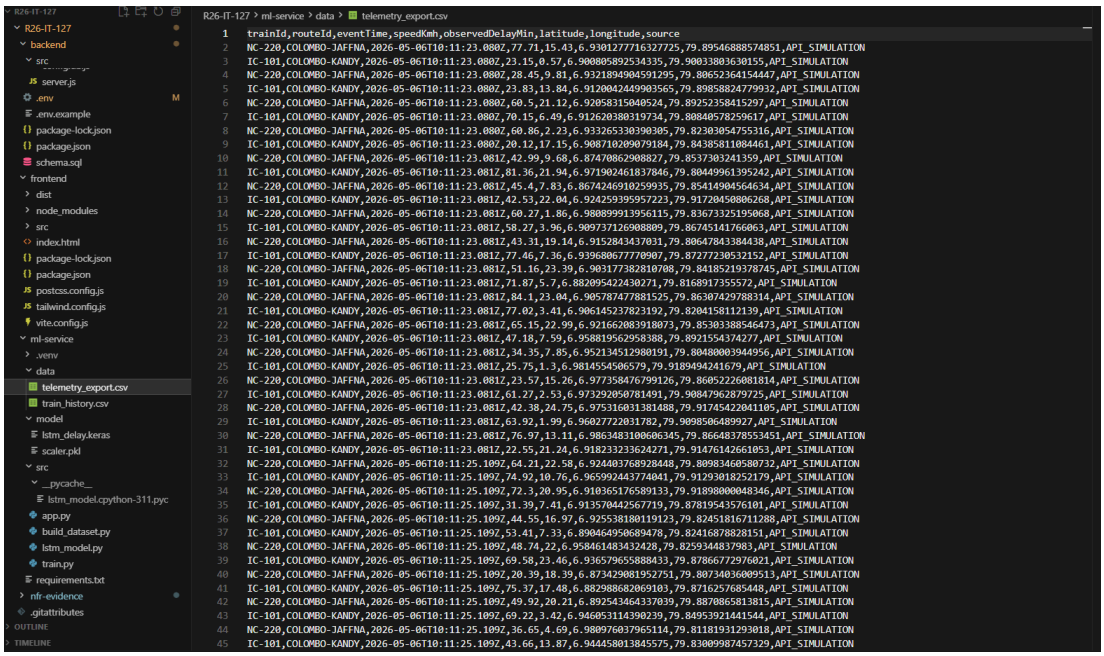


Figure 4 - Dataset Processing

Express delay consistently, such as in minutes. Define whether early arrivals (negative delays) are retained, clipped to zero, or analyzed separately, and apply the decision consistently. Feature engineering may include scheduled travel time, time/day, previous observed delay, elapsed journey time, and route segment, but only when source fields support them. Fit transformations such as scaling and imputation on training data only, then apply them to validation and test data to reduce leakage.

Stage 4: Tracking module development

Process available location or operational updates and associate them with the correct train and route. The interface may show the latest known position, update timestamp, route/station information, and feed status. Clearly identify stale or unavailable updates. Historical or simulated coordinates must be labelled as such.

Stage 5: Delay prediction model development

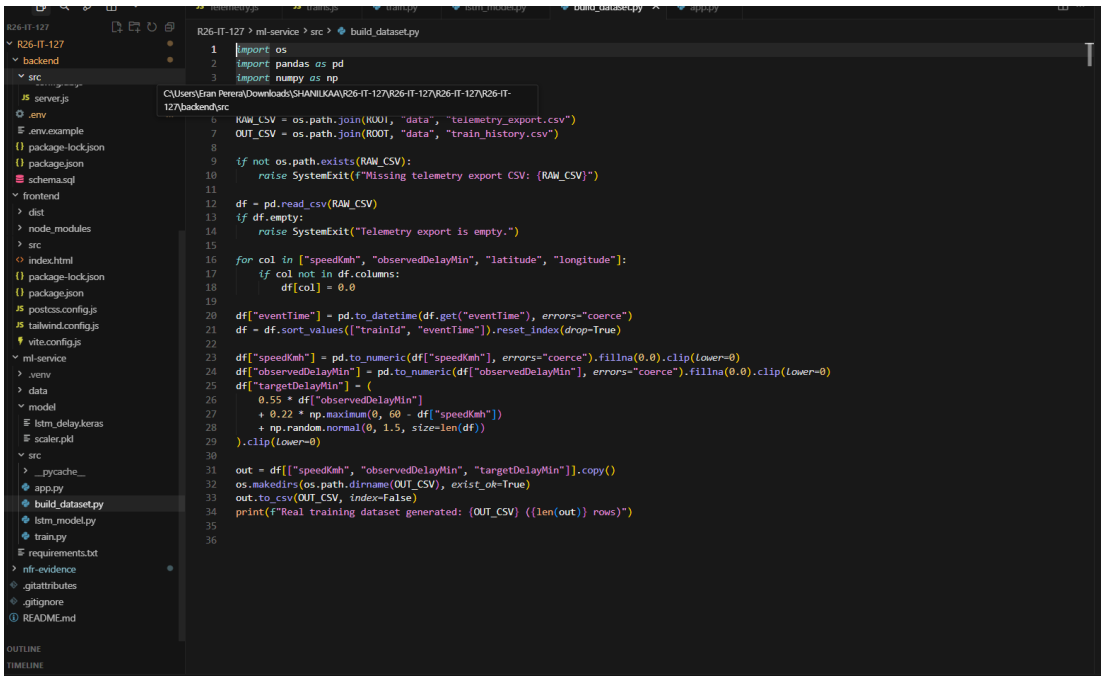


Figure 5 - Model Development

Define the prediction target before training. Targets may be arrival time at a specified station, delay in minutes, or a delay category (low, moderate, high); these are distinct tasks. Regression predicts a numerical value, while classification predicts a category. Define and justify category thresholds. Establish a baseline, such as scheduled arrival, the latest observed delay, or historical average travel time. Compare more complex models against it. Candidate models include linear regression, decision trees, random forests, gradient boosting, and sequence models when sufficient ordered time-series data exist. Present only models actually implemented as experimental results; others are alternatives.

Stage 6: Training, validation, and testing

Separate data into training, validation, and test sets. For time-dependent data, chronological splitting better reflects future prediction. Earlier journeys may be used for training, a later period for validation, and the latest held-out period for testing. Keep records from the same journey within one split. Select features and hyperparameters using training/validation data, reserve the test set for final evaluation, and calculate test metrics only after finalizing the model.

Stage 7: System integration and testing

Integrate tracking and prediction into the application. Verify correct train association, timestamp handling, station/journey alignment of predictions, and handling of missing or delayed updates. Test normal and edge cases, including missing locations, duplicate updates, unexpected station records, and insufficient history.

Stage 8: Evaluation and refinement

Evaluate using Section 3.4 metrics, compare with the baseline, and analyses prediction errors. Document changes after initial evaluation and use the designated held-out test data for final evaluation rather than repeatedly tuning against it.

3.4 Measures and evaluation metrics

Evaluate four areas: numerical delay prediction, classification where applicable, tracking, and system behavior. The selected measures depend on the implemented task.

3.4.1 Regression metrics for delay prediction

For numerical arrival-time or delay predictions, use:

Mean Absolute Error (MAE):

$$MAE = (1/n) \Sigma |y_i - \hat{y}_i|$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{[(1/n) \Sigma (y_i - \hat{y}_i)^2]}$$

Here, y_i is the observed value, $\hat{y}_i$ the prediction, and n the number of observations. If delay is measured in minutes, both metrics are in minutes. MAE indicates average absolute error; RMSE penalizes large errors more strongly.

Mean Absolute Percentage Error (MAPE) may be used when percentage error is meaningful:

$$MAPE = (100/n) \Sigma |(y_i - \hat{y}_i) / y_i|$$

MAPE is undefined or unstable when actual values are zero or close to zero. Since delays may be zero or negative, it should not be the principal metric for raw delay unless this issue is addressed. MAE and RMSE are generally more suitable. Compare

models on the same held-out observations and state whether the target is delay duration or absolute arrival time.

3.4.2 Classification metrics for delay categories

For categories such as low, moderate, and high, use a confusion matrix and:

- Accuracy: proportion of correctly classified observations.

- Precision: proportion of predictions for a class that are correct.

- Recall: proportion of actual class observations correctly identified.

- F1-score: harmonic mean of precision and recall.

- Macro-F1: mean F1 across classes, weighting each class equally.

Accuracy can be misled when classes are imbalanced. Review the confusion matrix and report justified class thresholds. Use regression metrics for numerical outputs and classification metrics for categories; report each separately if both are implemented.

3.4.3 Tracking performance measures

Table 3 - Tracking performance measures

Measure	Meaning
Location error	Difference between the reported position and a trusted reference position, if one is available
Update latency	Time between a location event occurring and the update becoming available to the system
Update frequency	Number of valid updates received within a defined time interval
Data availability	Proportion of the expected observation period during which usable tracking information is available
Staleness	Age of the most recently displayed update at the time it is viewed

Calculate location error only with a suitable reference position; latency requires appropriate timestamps. Historical station records can support evaluation of record processing but not claims of measured live GPS accuracy.

3.4.4 System-level performance

Where relevant, assess response time, successful processing rate, error handling, and interface usability. Measure server and client performance separately if appropriate, and record hardware, software environment, test duration, and measurement procedure. Define acceptance criteria before final testing where practical, such as maximum update age or acceptable prediction error. Justify thresholds using requirements and evidence, not merely observed results.

3.5 Data analysis

Analysis includes descriptive examination, target preparation, model comparison, error analysis, and interpretation.

3.5.1 Descriptive analysis

Summarize dataset size, coverage, and quality: number of journeys, station events and location updates; time period; delay distribution; and missing/invalid values. Mean, median, standard deviation, minimum, maximum, percentiles, histograms, box plots, and time-based plots can show typical delays and extreme cases.

3.5.2 Data quality analysis

Record the number and proportion of missing values, duplicate events, inconsistent timestamps, and implausible observations, along with actions taken. Do not automatically remove outliers: large delays may represent genuine disruptions. Correct or exclude records only under documented data-quality rules.

3.5.3 Model training and comparison

Train baseline and candidate models on training data, select models using validation data, and evaluate on the held-out test set. Compare regression models mainly by MAE and RMSE. For classification, report the confusion matrix, class-level precision and recall, macro-F1, and accuracy. Tables should identify each model, target, test set, and metric. If data permit, analyze results by route, station, time period, or delay range; interpret small subgroups cautiously.

3.5.4 Error analysis

For observation i, the residual is:

$$e_i = y_i - \hat{y}_i$$

For arrival-time prediction, the residual sign indicates whether the prediction was earlier or later under this definition; state the convention consistently. Examine

residuals against actual delay, scheduled time, route segment, and prediction horizon where available to identify systematic errors. For classification, inspect commonly confused classes and class-level recall, since high overall accuracy may conceal failure to identify less common classes.

3.5.5 Interpretation and reporting

Interpret findings against the research objectives, baseline improvement, operational importance of errors, data adequacy, and computational resources—not just the best single score. Report only results from actual experiments. If reference data or observations are insufficient, mark the relevant measure as unavailable rather than inventing a value. Discuss limitations from data coverage, missing fields, and offline testing.

3.6 Ethics, privacy, security and responsible AI

Consider privacy, tracking-data security, and the consequences of inaccurate predictions. Safeguards must match the data sources and deployment environment.

3.6.1 Ethical considerations

Use authorized data and comply with provider conditions on access, storage, analysis, publication, and sharing. For surveys, interviews, or usability tests, inform participants, obtain consent where required, ensure voluntary participation, and secure institutional ethical approval before recruitment when applicable. If only public or authorized datasets are used without direct participants, state this accurately; privacy, licensing, and responsible handling still apply.

3.6.2 Privacy and data minimization

Collect and retain only data needed for tracking, prediction, and evaluation. Avoid passenger names, phone numbers, account details, and other personal identifiers unless essential to a defined research function. Remove, anonymize, or pseudonymize personal information where appropriate and restrict access to remaining identifiers. Operational train-location data may also be sensitive; public display must comply with provider permissions and railway restrictions.

3.6.3 Data and application security

Use proportionate controls against unauthorized access, alteration, and disclosure, including:

- Authentication and role-based access for restricted functions.

- Secure data transmission where supported.

- Protection of credentials and configuration secrets.

- Input validation and handling of malformed records.

- Backups and controlled retention of research artefacts.

- Error logging without unnecessary personal data.

Manage database/API permissions and credentials separately from public source code. Describe security testing as comprehensive only if a full audit was actually conducted.

3.6.4 Responsible AI and prediction transparency

Predictions are estimates, not guarantees. Missing updates, schedule changes, disruptions, and unrepresentative training data can reduce accuracy. The system should:

- Distinguish scheduled, observed, and predicted times.

- Show the latest-update time when tracking data are used.

- Avoid presenting stale or unavailable information as current.

- Communicate uncertainty or limitations where meaningfully estimated.

- Evaluate performance across delay categories and operating conditions.

- Avoid claims that the model eliminates delays or guarantees arrivals.

Document model inputs, target definition, data splitting, and metrics. Do not use information unavailable at prediction time; for example, using the actual arrival time to predict that same arrival causes data leakage and misleading results.

3.6.5 Limitations and accountability

Treat the system as an information-support tool, not a replacement for authorized railway decisions. Document failure cases such as interrupted location feeds or insufficient prediction history and communicate them clearly where the design permits.

3.7 Reproducibility and research artefacts

Preserve sufficient information about data, software, configuration, and evaluation so that the work can be inspected and repeated.

3.7.1 Dataset documentation

Record each dataset's source, format, collection/download date, time coverage, variables, units, and limitations, along with cleaning rules, exclusions, feature transformations, and target definitions. If redistribution is restricted, explain access limitations and provide a permitted data dictionary or non-sensitive sample. Do not publish confidential railway data.

3.7.2 Source code and software configuration

Organize pre-processing, feature engineering, training, prediction, and evaluation code into clear modules where practical; document tracking and data ingestion too. Record:

- Programming languages and versions.

- Main libraries and versions.

- Database, framework, or mapping service, if used.

- Model type and hyperparameters.

- Feature definitions and target calculation.

- Data-splitting method and random seeds, where relevant.

- Setup and execution instructions.

If version control is used, record the commit ID or release tag associated with the experiments.

3.7.3 Model and experiment artefacts

Retain model configuration and reproducibility artefacts, such as trained models, preprocessing settings, feature lists, prediction outputs, evaluation scripts, confusion matrices, residual analyses, and metric summaries. Link each result to its model and dataset versions, split definition, and evaluation script. Record random seeds where applicable.

3.7.4 Testing and system documentation

Keep test cases and observed outcomes for tracking and prediction, including:

- Correct association of trains with location/station records.

- Correct delay calculation from scheduled and actual times.

- Handling of missing or duplicate records.

- Correct display of predicted arrival times and delays.

- Handling unavailable or outdated updates.

- Consistency between evaluation scripts and reported metrics.

Architecture and data-flow diagrams, database schemas where applicable, and prototype run instructions further support reproducibility.

4: PRODUCT AND SOLUTION DESIGN

4.1 Introduction

This chapter presents the design of the Real-Time Train Tracking and Delay Prediction component within the wider Sri Lankan railway information system. It covers railway-data preparation, display of the latest available movement information, delay prediction, and evaluation of system performance and limitations.

The component combines tracking, which reports a train's latest recorded position or movement status, with prediction, which estimates arrival time or delay from available operational data. A modular design separates data acquisition, preprocessing, tracking, prediction, storage, and presentation to support testing, maintenance, and future improvement. Implementation depends on data quality and availability, selected models, and development resources.

4.1 Requirements

Requirements define the functions and quality characteristics needed to meet the research objectives. They support data preparation, tracking, prediction, model evaluation, integration testing, and future improvement. Functional requirements specify system operations; non-functional requirements describe quality, reliability, usability, and security.

4.1.1 Functional requirements

Table 4 - Functional requirements of the proposed system

ID	Functional requirement	Description	Related objective
FR-01	Railway data acquisition	The system shall support obtaining available timetable, train identifier, route, station, timestamp, location, and delay information from the configured data sources.	Objective 1
FR-02	Data validation	The system shall identify missing fields, invalid values, duplicate records, and inconsistent timestamps before the data are used.	Objective 2
FR-03	Data alignment and feature preparation	The system shall align available train identifiers, location observations, station records, and timestamps, and	Objective 2

		prepare relevant features for tracking and prediction.	
FR-04	Train tracking	The system shall present the latest available train position or movement status according to the information supported by the data source.	Objective 3
FR-05	Tracking freshness and source indication	The system shall display the timestamp and, where available, the source of the tracking information so that users can assess its freshness.	Objective 3
FR-06	Delay calculation and target preparation	The system shall calculate observed delays or prepare the appropriate prediction target from available scheduled and actual event times.	Objectives 1 and 4
FR-07	Delay-prediction model execution	The system shall support the selected and implemented prediction model or models using the prepared input features.	Objective 4
FR-08	Prediction presentation	The system shall present the predicted arrival time, delay duration, or delay category according to the implemented prediction task.	Objective 4
FR-09	Prediction performance evaluation	The system shall support evaluation using appropriate metrics, including MAE and RMSE for regression or precision, recall, F1-score, accuracy, and confusion matrices for classification.	Objective 5
FR-10	Model comparison	Where multiple models are implemented, the system shall support comparing their performance using consistent evaluation data and metrics.	Objective 5
FR-11	Component integration	The system shall integrate the available data-processing, tracking, prediction, and presentation components so that their outputs can be associated with the correct train and journey.	Objective 6
FR-12	Testing and error handling	The system shall support testing of data processing, tracking, prediction, and integration, including handling	Objective 6

		unavailable, invalid, or insufficient data.	
FR-13	Evaluation and limitation reporting	The research shall document observed limitations relating to data availability, prediction performance, update latency, and deployment readiness.	Objective 6
FR-14	Future improvement recommendations	The evaluation shall identify evidence-based opportunities to improve reliability, uncertainty communication, and suitability for future railway use.	Objective 7

The requirements cover the full workflow, not just the interface. FR-02 and FR-03 support data quality, which affects prediction reliability; FR-09 and FR-10 ensure model choices rely on measured performance. Linear Regression, Random Forest Regression, and Long Short-Term Memory (LSTM) may be considered, but only implemented and evaluated models should be reported as completed work.

4.1.2 Non-functional requirements

Table 5 - Non-functional requirements

ID	Requirement	Description
NFR-01	Performance	Data processing, prediction execution, and result presentation should meet the needs of the intended application.
NFR-02	Reliability	The system should handle missing records, invalid inputs, and unavailable data sources without presenting unsupported information as valid.
NFR-03	Prediction quality	Prediction performance should be measured against observed outcomes using appropriate metrics and a separate test dataset.
NFR-04	Usability	Train status, timestamps, scheduled arrivals, predicted arrivals, and delay values should be clearly labelled and understandable.
NFR-05	Maintainability	The data-processing, tracking, prediction, and presentation functions should be separated logically to simplify testing and future changes.
NFR-06	Scalability	The design should allow additional train records, routes, stations, or data sources to be accommodated where resources permit.

NFR-07	Security and privacy	Access to restricted data and credentials should be controlled, and unnecessary personal information should not be collected.
NFR-08	Reproducibility	Data preparation, model configuration, and evaluation procedures should be documented sufficiently to support verification of results.
NFR-09	Transparency	The interface should distinguish actual observations from predictions and identify stale or unavailable tracking information where possible.
NFR-10	Compatibility	The components should operate within the selected software environment and exchange data using consistent formats.

These requirements guide testing. Reliability can be tested with missing or invalid records, prediction quality with appropriate metrics, and performance/freshness through measurements under a defined setup rather than unsupported claims.

4.1.3 Constraints and assumptions

Official timetables, actual arrival records, GPS coordinates, and live feeds may not all be available. Geographic tracking requires valid location data; with station events only, the system should show the latest recorded station/status, not claim continuous geographic tracking. Prediction quality depends on training-data quality, volume, and coverage; limited history may weaken generalization across routes, seasons, and unusual conditions.

The design assumes records can be associated with the relevant train and, where applicable, journey and station, and that timestamps and scheduled/actual event times are consistently interpretable. Any unmet assumption must be documented and addressed through validation or alternative rules.

4.2 Component architecture within the complete solution

4.2.1 Architectural overview

The modular, data-driven architecture processes railway data before passing them to tracking and delay prediction, then presents results for users and evaluation. The component converts available operational data into movement information and delay estimates; it does not control railway operations or guarantee arrival times.

The architecture has five logical layers:

1. Data source: timetables, historical journeys, station events, and location observations where available.

2. Data processing: validation, standardization, train/timestamp alignment, and feature preparation.

3. Tracking and prediction: latest movement status and execution of the selected prediction model.

4. Storage: schedules, processed records, tracking observations, model artefacts, and predictions as required.

5. Presentation and evaluation: display results and assess system/model performance.

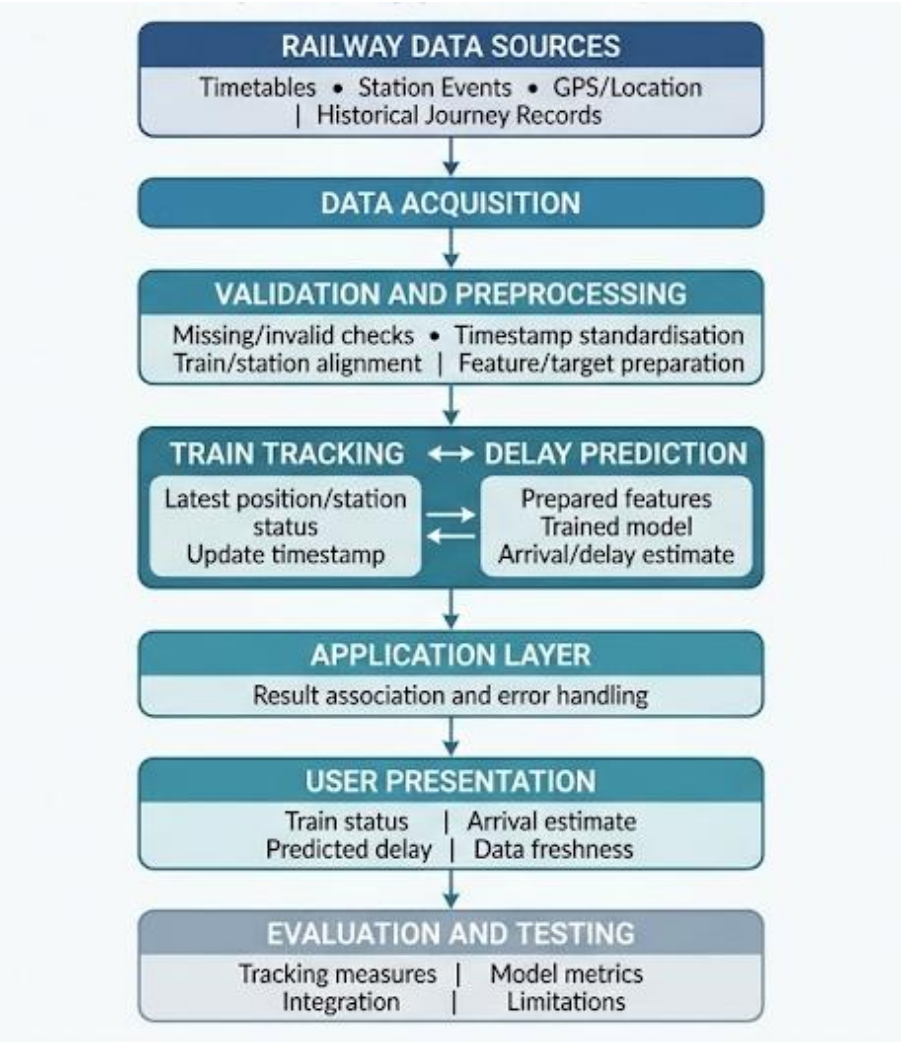


Figure 6 - component Architecture

The layers may be modules within one application rather than separately deployed services. Storage may support several workflow stages, and tracking can still be displayed when prediction is unavailable. The modular structure allows preprocessing to be tested with known records, tracking against suitable reference observations, and prediction against held-out data before integration.

4.2.2 Data acquisition and processing

Data may come from files, a database, an authorized API, or a location feed. Each record should retain identifiers and timestamps needed to associate it with the correct train and event. Preprocessing standardizes timestamps, detects duplicates, validates station IDs, and derives delay features from scheduled and actual times where possible.

Prediction inputs must be available at the time a real prediction is made. Using information that becomes available only after the target arrival event introduces data leakage and misleading results.

4.2.3 Train tracking component

The tracking component receives validated location or station-event records and identifies each train's latest available status. Coordinates can support a map view; station events can support the latest recorded station/event only. Output must reflect the actual data source. Display the update timestamp and mark stale information; missing updates do not prove that a train has stopped.

4.2.4 Delay-prediction component

The component uses prepared operational features to predict delay in minutes, arrival time at a specified station, or—if classification is implemented—a defined delay category. Train on historical observations and evaluate with data excluded from model selection. Apply the same preprocessing and features at inference, associate each estimate with the correct train and prediction time, and label it clearly. Distinguish predictions from scheduled and actual arrivals. If inputs are insufficient, return an unavailable status or a clearly identified fallback only if implemented.

4.2.5 Storage and presentation

Storage may use structured files, a relational database, or another suitable mechanism. The interface may display train ID, latest reported position/station, update time, scheduled arrival, predicted arrival, and estimated delay. Clearly identify the data

source and freshness. Label historical or simulated data when live location data are unavailable.

4.3 Design alternatives and decisions

Alternatives are assessed by data availability, complexity, prediction suitability, maintainability, and evaluability. The aim is to meet research objectives without unnecessary complexity.

4.3.1 Train-tracking alternatives

GPS-based tracking uses coordinates from a GPS device or authorized feed and supports map-based updates, but depends on location data, communication, and possibly hardware. Station-event tracking represents progress through arrival/departure events but cannot establish exact position between stations. Historical/simulated tracking reconstructs movement for repeatable development and testing but cannot prove live performance.

Choose according to available data: GPS when valid coordinates exist, station-event tracking when only station records are available, and historical/simulated tracking for prototypes when limitations are clearly disclosed.

4.3.2 Delay-prediction alternatives

Table 6 - Comparison of candidate delay-prediction approaches

Approach	Advantages	Limitations	Potential role
Timetable-based estimate	Simple and transparent	Does not independently account for developing delays	Baseline
Historical average	Easy to interpret and implement	May not adapt to changing conditions	Baseline or data-limited approach
Linear Regression	Simple numerical model with interpretable relationships	May not represent complex nonlinear patterns	Regression baseline
Random Forest Regression	Can model nonlinear relationships in tabular features	Requires suitable data and careful validation	Candidate regression model

LSTM	Can learn temporal patterns from ordered sequences	Requires suitable sequential data and additional training complexity	Candidate time-series model

Timetable or historical-average estimates provide baselines. Linear Regression offers a simple regression reference; Random Forest may capture nonlinear relationships; LSTM may suit sufficient sequential data where temporal dependencies matter. These are alternatives, not mandatory implementations. Report only models actually trained and evaluated, selecting based on available data and defined metrics.

4.3.3 Application architecture alternatives

A monolithic architecture puts all functions in one application, simplifying initial development but making independent replacement harder. A modular architecture separates processing, tracking, prediction, and presentation for independent testing and maintenance without requiring separate deployments. A distributed architecture deploys separate services and may scale independently, but adds deployment, communication, and monitoring complexity.

A modular architecture suits the distinct tracking, prediction, and evaluation objectives. Modules may remain in one application or be deployed separately if justified. A distributed architecture is unnecessary unless actual requirements warrant the complexity.

4.3.4 Data-storage alternatives

File-based storage may suit small datasets and repeatable offline experiments. Relational databases better support relationships among trains, stations, schedules, events, and predictions. Other mechanisms may suit high-volume or streaming needs. Select based on data size, relationships, update frequency, effort, and persistence needs, and document the storage actually implemented.

4.3.5 Summary of design decisions

The design separates tracking and prediction, uses transparent data preparation, establishes a simple prediction baseline, and selects technologies suited to available data and resources. The final thesis should document chosen technologies, alternatives, and evidence for decisions, distinguishing actual implementation from candidate options.

4.4 Detailed design

This section describes the inputs, processing, outputs, and interactions of principal components to support implementation, testing, and evaluation.

4.4.1 Data-processing design

The data-processing module converts raw railway records into a consistent form for tracking and prediction. Its workflow is:

1. Acquire/load available records.
2. Validate required fields, train/station IDs, and timestamps.
3. Standardize data types, time formats, and units.
4. Identify duplicates, missing values, and invalid records.
5. Align records to train, journey, and station where possible.
6. Derive model features and targets.
7. Supply validated records to tracking and prediction.
8. Record data-quality issues and processing errors.

For a station-arrival event:

$$D = T_a - T_s$$

where D is observed delay, T_a actual arrival time, and T_s scheduled arrival time. Express the difference in a consistent unit, such as minutes. Retain negative early-arrival values or apply a documented rule consistently; missing actual arrivals must not be treated as zero delay.

Features may include time/day, scheduled travel time, previous observed delay, elapsed journey time, and route/station information where available. Fit data-dependent preprocessing on training data, then apply it to validation/test data to prevent leakage.

4.4.2 Train-tracking design

The tracking module selects the latest accepted location or movement record and supplies it to the interface:

1. Receive a validated location update or station event.
2. Identify the train and journey.

3. Validate timestamp and position/station information.
4. Compare with the latest accepted update.
5. Update status only if validation rules are satisfied.
6. Send status, source, and timestamp to the interface.

Where available, coordinate records may include train ID, latitude, longitude, timestamp, and source for map markers and update times. With station events only, show the latest station and event time; do not infer exact between-station position without an explicit estimation method. Handle old and duplicate updates consistently so an older record does not replace a newer valid one. If no reliable update exists, indicate that current information is unavailable.

4.4.3 Delay-prediction design

The prediction workflow is:

1. Receive a valid request and input features.
2. Check required features.
3. Apply training-time preprocessing.
4. Execute the selected trained model.
5. Validate the output and convert units.
6. Associate the result with the train, target station/event, and prediction time.
7. Return the estimate to the application.

If predicting delay, arrival time may be calculated as:

$$\hat{T}_a = T_s + \hat{D}$$

where $\hat{T}_a$ is predicted arrival, T_s scheduled arrival, and $\hat{D}$ predicted delay. They must refer to the same event and use compatible units/definitions. If the model directly predicts arrival time, do not add another delay estimate. A delay-category classifier does not produce an exact arrival time unless a separate defined calculation supports it. Keep feature order, transformations, and model configuration consistent with training; handle missing inputs and invalid outputs without presenting arbitrary values as valid predictions.

4.4.4 Data-model design

The data model should preserve consistent associations between trains, stations, schedules, observed events, location updates, and predictions.

Table 4.4: Conceptual data entities

Table 7 - Conceptual data entities

Entity	Example attributes	Purpose
Train	Train identifier, service identifier	Identifies a train or service
Station	Station identifier, station name, coordinates if available	Identifies a railway station
Schedule	Train identifier, station identifier, scheduled event time	Stores planned movement information
Journey event	Train identifier, station identifier, actual event time, event type	Records observed arrivals or departures
Location update	Train identifier, latitude, longitude, timestamp, source	Records geographic observations where available
Prediction	Train identifier, station identifier, predicted value, prediction time, model version	Records prediction outputs

This model may use relational tables, structured files, or another store. Primary/foreign keys can protect integrity and indexes improve queries in relational databases. The actual schema must match implementation; offline experiments may use one prepared dataset, while the conceptual entities clarify data meaning and relationships.

4.4.5 User-interface design

The interface should quickly present:

- Train/service ID.
- Latest reported position or station status.
- Update timestamp and source, where available.
- Scheduled arrival at the selected station.
- Predicted arrival or estimated delay, according to the task.
- Clear messages for unavailable, stale, or insufficient data.

Explicitly distinguish scheduled, observed, and predicted values; use consistent delay units and timestamp formats. Map markers must represent the location associated with the displayed update. Animation must not imply continuous tracking unless supported, and historical/simulated information must be labelled. Communicate status through text and labels as well as colour for users with colour-vision difficulties.

4.4.6 Integration and error-handling design

The integration layer associates processing, tracking, and prediction outputs with the correct train, journey, and station. Define handling for:

- Missing/invalid train IDs.
- Incorrect/inconsistent timestamps.
- Duplicate or out-of-order updates.
- Unavailable data sources.
- Missing prediction features.
- Failed model execution or invalid outputs.
- Outdated tracking information.

Reject or flag invalid records under documented rules. Do not display failed predictions as valid numbers. If the feed is unavailable, show the last-known status with timestamp and freshness warning where possible. Record errors for debugging without exposing sensitive data and test each important failure case.

4.4.7 Testing and evaluation design

Testing determines whether the implementation meets its requirements:

- Unit testing: timestamp conversion, delay calculation, validation, and feature preparation.
- Integration testing: correct data exchange among tracking, prediction, storage, and presentation; predictions must be shown against the correct train.
- System testing: functional/non-functional requirements, error handling, freshness, and response time where relevant.
- Model evaluation: held-out observations, using MAE/RMSE for regression or accuracy, precision, recall, F1-score, and confusion matrix for classification.

Regression metrics:

$$MAE = (1/n) \Sigma |y_i - \hat{y}_i|$$

$$RMSE = \sqrt{[(1/n) \Sigma (y_i - \hat{y}_i)^2]}$$

where y_i is observed, $\hat{y}_i$ predicted, and n the number of test observations. Compare models with a baseline on the same test data. Chronological splitting is generally suitable for time-dependent data; prevent records from one journey leaking across splits when this reveals test outcomes.

Tracking evaluation requires reference data: location error needs a trusted position; latency needs event and receipt timestamps; availability needs a defined observation period and expected update behavior. Document test cases, expected and observed results, and unresolved issues. Report software correctness separately from prediction accuracy, since a correct application may still use an inaccurate model.

4.5 Standards and professional practice

Recognized software-engineering and responsible data-management practices guide quality, privacy, usability, and model evaluation. Relevance depends on implementation; discussing a standard does not prove formal compliance or certification.

4.5.1 Software quality and engineering practice

ISO/IEC 25010 provides a reference for functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and flexibility. Translate relevant characteristics into testable requirements: for example, test reliability with missing data, usability through interface tests, and efficiency through measured processing/response times.

Use modular code, consistent naming, documented interfaces, and appropriate error handling. Trace requirements to test cases so evaluation shows how the component was assessed.

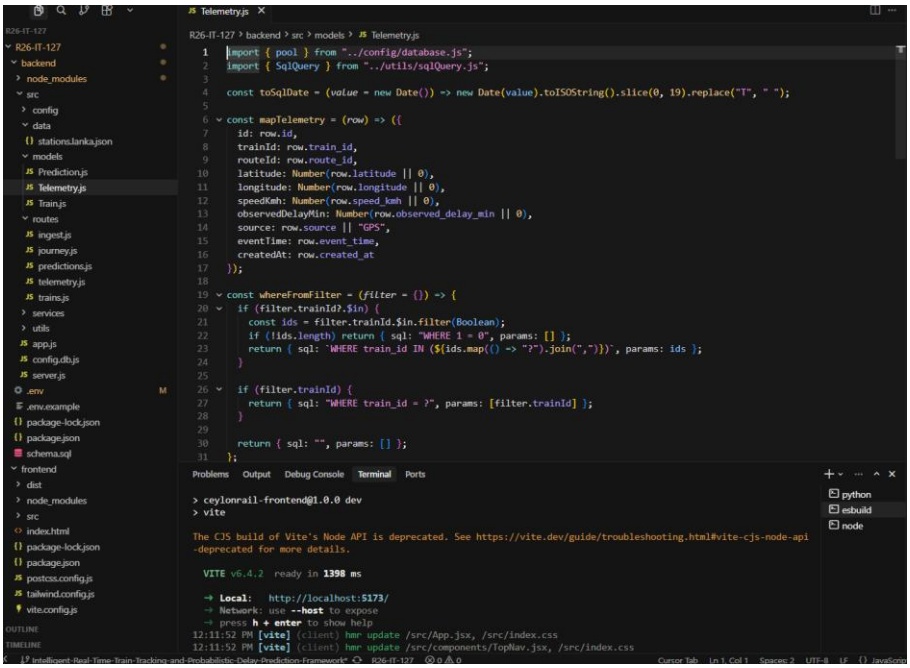


Figure 7 - code Snippet

4.5.2 Data quality and management

Data quality affects tracking and prediction. Document dataset sources, meaning, units, time coverage, and limitations. Apply and record cleaning, feature transformations, exclusions, and delay calculations consistently. Follow provider conditions for storage and sharing; do not publish restricted operational data without authorization.

4.5.3 Security and privacy

Apply suitable access controls and protect credentials and restricted data. Use secure network communication where supported, validate inputs, and avoid unnecessary personal/confidential information in logs. Collect only data needed for tracking, prediction, and evaluation; consider minimization, anonymization, or pseudonymization where personal data exist.

Security testing should match project scope. Do not describe an offline prototype as a production-secure railway service unless its controls and deployment have been assessed.

4.5.4 Accessibility and interface design

Use clear labels, readable text, consistent units, and a clear information hierarchy. Do not rely on colour alone for status differences; explain errors plainly. Distinguish scheduled, actual, and predicted arrival times and update timestamps to reduce ambiguity.

4.5.5 Responsible machine learning

Document target, input features, model configuration, data split, and metrics. Prevent leakage by excluding information unavailable at prediction time and keeping the test set separate from model selection and tuning. Use task-appropriate measures; for imbalanced delay categories, include class-level measures, while MAE/RMSE communicate numerical error.

Present outputs as estimates, not guarantees. Show uncertainty only when meaningfully estimable, and discuss limitations during disruptions or underrepresented operating conditions.

4.5.6 Documentation and reproducibility

Document target, input features, model configuration, data split, and metrics. Prevent leakage by excluding information unavailable at prediction time and keeping the test set separate from model selection and tuning. Use task-appropriate measures; for imbalanced delay categories, include class-level measures, while MAE/RMSE communicate numerical error.

Present outputs as estimates, not guarantees. Show uncertainty only when meaningfully estimable, and discuss limitations during disruptions or underrepresented operating conditions.

4.5.7 Professional responsibility and operational limitations

Tracking and predictions can affect passenger expectations and travel decisions. Distinguish observations from estimates and never present stale information as current. Do not describe a prototype as an official railway service without authorization and deployment. Do not generalize limited historical-data results to all Sri Lankan railway services without evidence.

Report limitations in data availability, location accuracy, latency, model performance, and deployment. These guide realistic improvements and clarify the

5. IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The Real-Time Train Tracking and Delay Prediction system combines railway data preparation, train-status processing, delay prediction, and model evaluation to support train monitoring and estimated delays. Development involved preparing the dataset, processing features, implementing the prediction model, integrating the workflow, and evaluating performance, with attention to data consistency, reliability, and interpretability.

5.1.1 Programming Language and Development Tools

Python was the primary language for structured-data processing, numerical computation, model training, and evaluation. Data-processing libraries supported data inspection and preparation; machine-learning tools supported model implementation and metric calculation. LSTM was used for the regression experiment because it can learn sequential relationships from appropriately prepared time-dependent data.

Table 5.1: Development technologies and their purposes

Table 8 - Development technologies and their purposes

Technology or tool	Purpose
Python	Implementing data processing, model execution, and evaluation
Data-processing libraries	Loading, cleaning, transforming, and preparing railway datasets
Machine learning libraries	Implementing predictive models and calculating evaluation metrics
LSTM model framework	Developing the sequence-based regression model for delay prediction
Data visualization tools	Presenting model performance through classification reports, confusion matrices, and regression metrics
Development environment	Writing, executing, debugging, and testing the system

Separating data preparation, model execution, and evaluation supported testing and maintainability.

5.1.2 Data and Processing Environment

The workflow uses available train identifiers, station identifiers, scheduled and actual arrival times, route information, and delay records. Preparation checks missing or invalid records, standardizes timestamps, and ensures consistent train and station identifiers.

Train delay is calculated as:

$$\mathbf{D = T_a - T_s}$$

where D is observed delay, T_a is actual arrival time, and T_s is scheduled arrival time. Times must use consistent units, such as minutes; positive values indicate late arrivals and negative values indicate early arrivals.

Training and testing data were separated so performance could be measured on unseen observations. Chronological order is particularly important for time-dependent data because it better represents future prediction conditions.

5.2 Component Implementation

The system comprises connected stages for data preparation, train-status processing, delay prediction, and performance evaluation.

5.2.1 Data Loading and Preprocessing

The dataset was loaded and inspected to identify fields needed for tracking and prediction. Preprocessing addressed missing values, inconsistent timestamps, duplicates, and invalid formats. Train and station identifiers were standardized, and timestamps were normalized for delay calculation and temporal feature extraction.

Features such as scheduled times, travel durations, previous delays, and other available operational attributes were prepared for model execution. Validation and transformation reduce errors caused by invalid timestamps, incorrect record associations, or inconsistent input formats.

5.2.2 Train-Tracking Functionality

The tracking component associates movement observations with train identifiers, orders them by timestamp, and identifies each train's latest valid observation. Available output includes train identifier, route, last reported station or position, observation timestamp, and movement status.

Displaying timestamps communicates information freshness. Recorded locations are observations, whereas estimated arrival times are model predictions; this distinction prevents predictions from being mistaken for verified positions.

5.2.3 LSTM-Based Delay Prediction

The LSTM regression experiment estimates numerical train delays from prepared railway data. As a recurrent neural-network architecture, LSTM can learn temporal relationships in sequential data. Inputs were structured for the model, which was trained on the training dataset and evaluated on a separate test dataset against actual delay values.

Estimated arrival time is calculated as:

$$\hat{T}_a = T_s + \hat{D}$$

where $\hat{T}_a$ is estimated arrival time, T_s is scheduled arrival time, and $\hat{D}$ is predicted delay; time units must be compatible. The estimate is not guaranteed and depends on input quality and predictive performance. Regression metrics assess differences between predicted and actual delays. This experiment is distinct from delay classification.

5.2.4 Delay Classification

Classification Report for Delay Classes

class	precision	recall	f1-score	support
On-time / Low Delay	0.000	0.000	0.000	212
Moderate Delay	0.345	1.000	0.513	367
High Delay	0.000	0.000	0.000	486
accuracy			0.345	1065
macro avg	0.115	0.333	0.171	1065
weighted avg	0.119	0.345	0.177	1065

Figure 8 - Delay Classification Report

The classification experiment assigns observations to On-time / Low Delay, Moderate Delay, or High Delay. Predicted labels were compared with test labels using precision, recall, F1-score, accuracy, and confusion matrices.

Testing showed that every observation was assigned to Moderate Delay. Thus, although implemented and evaluated, the classifier currently cannot reliably distinguish the three categories.

5.3 Integration with the Other Components

Data processing supplies structured inputs to tracking and prediction. Integration maintains consistency across preparation, model execution, and presentation, particularly for train and station identifiers and timestamps.

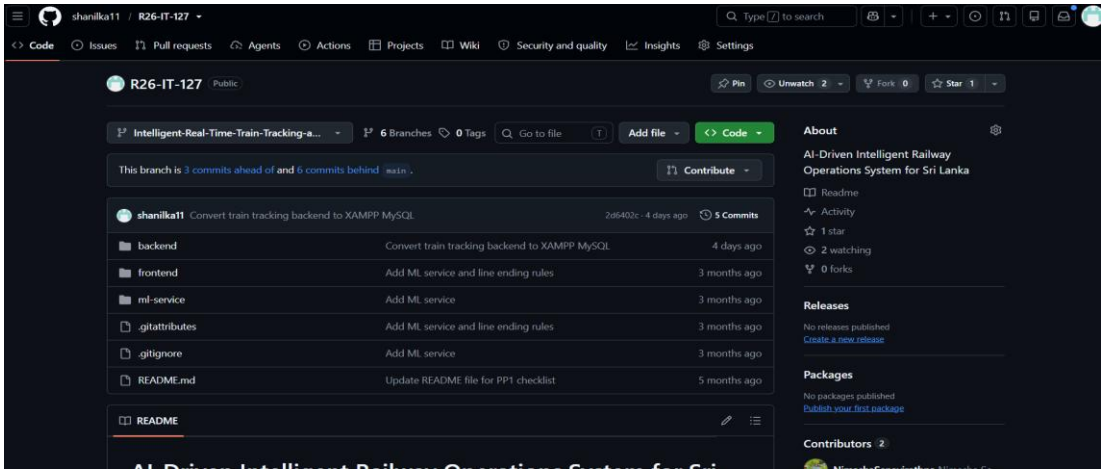


Figure 9 - GitHub I

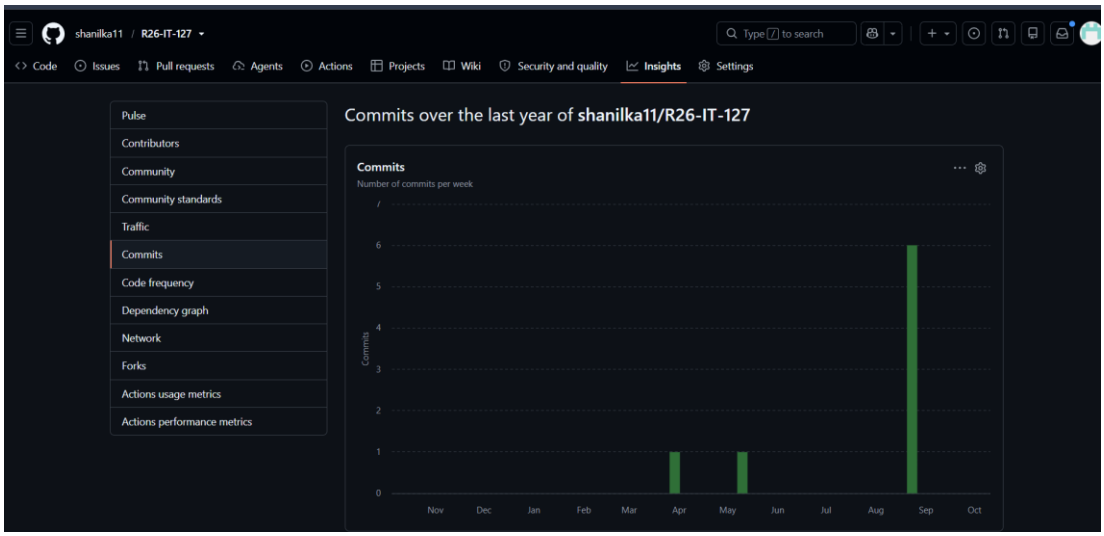


Figure 10 - GitHub II

5.3.1 Data-Processing Integration

The integrated workflow is:

Load available schedules, movement records, and historical delay observations.

Validate fields and standardize identifiers, timestamps, and numerical values.

Prepare features for tracking and prediction.

Identify each train's latest valid movement observation when available.

Run the trained model with appropriately prepared inputs.

Associate predictions with the relevant train and target station.

Present tracking information and prediction outputs for interpretation and evaluation.

This workflow reduces inconsistencies and helps locate errors when records cannot be processed or predictions cannot be generated.

5.3.2 Integration of Tracking and Prediction Outputs

Tracking provides the latest available movement observation; prediction estimates delay or arrival time. Outputs may include train identifier, last reported location or station, observation timestamp, predicted delay, and estimated arrival time when available.

Recorded and predicted information must remain distinct. An old location is a historical observation, not a confirmed current position. Predictions must also be associated with the correct train and target station; consistent identifiers and timestamps are essential.

5.3.3 Error Handling and Data Consistency

Inputs may be incomplete, contain invalid timestamps, or lack required features. The workflow checks suitability and handles invalid records according to preprocessing rules rather than silently creating valid-looking outputs. If required input is unavailable, no unsupported prediction should be presented.

Training and inference must use the same preprocessing configuration, feature order, transformations, scaling, and—where applicable—LSTM sequence structure. These controls reduce errors caused by mismatched formats or invalid values.

5.4 Verification and Software Testing

Testing examined the processing and prediction workflow, including classification and regression. Functional verification checks intended operations; model evaluation measures agreement between predictions and actual values or classes.

5.4.1 Functional Testing

Table 9 - Functional test cases

Test ID	Test case	Expected behavior
TC-01	Load railway dataset	The dataset is loaded and the required fields are accessible
TC-02	Validate train and station identifiers	Records are associated with the correct train and station
TC-03	Validate timestamps	Timestamp values are converted to a consistent format
TC-04	Prepare prediction inputs	Features match the input structure expected by the model
TC-05	Execute the LSTM regression model	A numerical delay prediction is generated for valid input
TC-06	Generate classification results	The classifier returns a predicted delay category
TC-07	Calculate classification metrics	Precision, recall, F1-score, and accuracy are calculated
TC-08	Generate confusion matrices	Actual and predicted classes are compared

TC-09	Evaluate regression predictions	Numerical predictions are compared against actual delay values
TC-10	Handle incomplete input	Invalid or insufficient input is handled without presenting an unsupported prediction
TC-11	Check tracking timestamps	The displayed tracking observation includes its associated timestamp

5.4.2 Classification Performance Evaluation

The classifier was evaluated on 1,065 test observations across On-time / Low Delay, Moderate Delay, and High Delay.

Table 10 - Classification report for delay classes

Delay class	Precision	Recall	F1-score	Support
On-time / Low Delay	0.000	0.000	0.000	212
Moderate Delay	0.345	1.000	0.513	367
High Delay	0.000	0.000	0.000	486
Macro average	0.115	0.333	0.171	1,065
Weighted average	0.119	0.345	0.177	1,065

Reported accuracy was approximately 34.5%. Moderate Delay recall was 1.000, while On-time / Low Delay and High Delay had zero precision, recall, and F1-score. The macro- and weighted-average F1-scores were 0.171 and 0.177, indicating weak performance. Per-class metrics are therefore necessary; accuracy alone is insufficient..

5.4.3 Confusion Matrix Analysis

The confusion matrix compares actual and predicted delay categories.

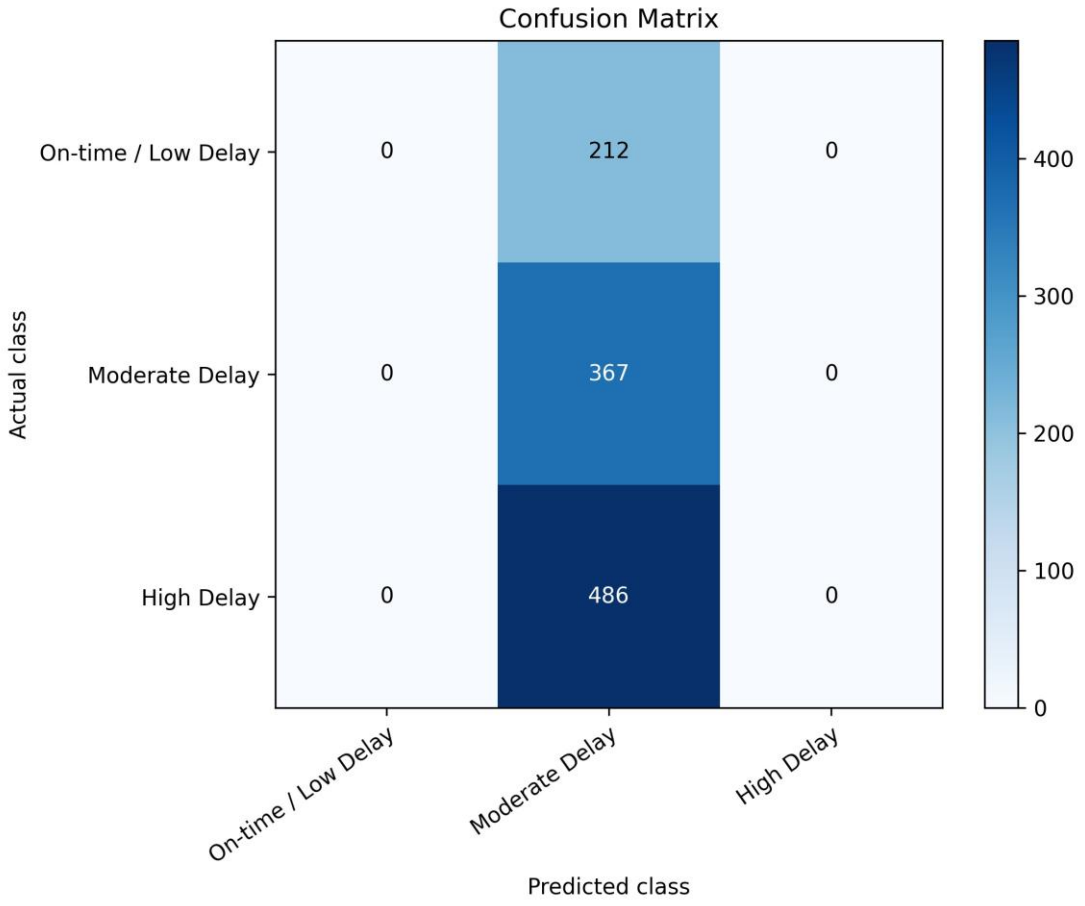


Figure 11 - Confusion Matrix

Table 11 - Confusion matrix for delay classification

Actual class	Predicted On-time / Low Delay	Predicted Moderate Delay	Predicted High Delay
On-time / Low Delay	0	212	0
Moderate Delay	0	367	0
High Delay	0	486	0

All 1,065 observations were predicted as Moderate Delay. Of these, 367 actual Moderate Delay observations were correctly classified and 698 were misclassified, giving:

Accuracy = 367 / 1065 ≈ 0.345

The normalized matrix shows 100% of each actual class assigned to the Moderate Delay column, with no predictions for the other classes. The classifier therefore fails to distinguish low/on-time, moderate, and high delays.

Possible causes to investigate include label preparation, training-class distribution, predicted-label encoding, and training configuration; the results do not confirm a single cause. The model requires refinement and retesting before supporting dependable class-specific alerts.

5.4.4 LSTM Regression Evaluation

Regression Metrics for LSTM Delay Prediction

model	test_samples	mae_minutes	mse	rmse_minutes	r2	classification_accuracy
lstm_delay	1065	4.092	24.859	4.986	0.001	0.345

Figure 12 - Regression Metrics for LSTM

The LSTM regression experiment predicts numerical delay values, unlike classification, which assigns predefined categories. Regression performance is assessed using measures including Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

$$\mathbf{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$\mathbf{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Here, y_i is the actual delay, $\hat{y}_i$ the predicted delay, and n the number of test observations. MAE gives the average absolute error; RMSE penalizes larger errors more strongly. If delay is measured in minutes, both metrics are in minutes.

Regression findings must be interpreted separately from the classification report: the 34.5% classification accuracy does not describe LSTM regression performance. MAE and RMSE should be considered together and compared with a suitable baseline.

5.4.5 Testing Findings and Identified Limitations

Testing established a workflow for evaluating both classification and regression, but revealed that the classifier predicts Moderate Delay for every test observation. Label

preparation, class distribution, feature processing, and model configuration require investigation. Regression must be assessed separately using numerical error metrics.

Tracking reliability also depends on fresh, consistent observations associated with the correct train. Train and station identifiers and timestamps must be preserved throughout processing. The results identify implemented capabilities and areas for further model refinement.

5.5 Deployment and Operation

The system combines railway data processing, train-status handling, delay prediction, and evaluation. Operation requires compatible inputs, processing components, and a trained model; training-time transformations and feature structures must be preserved during inference.

5.5.1 Deployment Environment

The software can run in a compatible Python environment with required data-processing and machine-learning libraries. The dataset, trained model, model files, and preprocessing configuration must be available. Adequate processing and storage resources are required.

Local or server deployment can support execution; cloud infrastructure may provide remote access and greater processing capacity. The choice depends on operational needs, resources, and intended users.

5.5.2 Operational Workflow

During operation, available railway records are processed for tracking and prediction. The latest valid movement observation is identified, and the model estimates delay and arrival time when inputs permit. Outputs can associate the train identifier and reported status with predicted delay and estimated arrival time; timestamps indicate tracking freshness.

Observed locations must remain distinct from predicted arrival times. Because the evaluated classifier predicts Moderate Delay for every test sample, its output cannot reliably distinguish the three categories and must be treated as a limitation, not dependable operational categorization.

5.5.3 Monitoring and Maintenance

Ongoing monitoring should check data completeness, invalid timestamps, processing errors, and expected feature formats. Once actual arrival information is available, regression should be assessed with MAE and RMSE; classification should be reviewed using precision, recall, F1-score, accuracy, and confusion matrices.

After changes to data preparation or training, the classifier should be reassessed on a separate test dataset. Maintenance should preserve model versions and preprocessing configurations; feature or format changes require pipeline updates and retesting.

5.5.4 Reliability, Security, and Future Improvements

Reliable outputs depend on data quality and correct interpretation. The system should retain observation timestamps, indicate stale or unavailable information, and avoid presenting predictions as confirmed facts. Access to sensitive operational data should be appropriately restricted, and unnecessary personal information excluded.

The classifier's single-class behavior should be investigated through checks of target labels, class distribution, feature preparation, and training configuration, followed by retraining and evaluation across all three classes. Further work may compare classification models, tune parameters, use baselines, and examine additional observations. Regression should also be tested under different temporal and operational conditions..

6. RESULTS AND VALIDATION

6.1 Research Results

The research developed and evaluated a Real-Time Train Tracking and Delay Prediction system integrating railway data processing, train-status handling, machine-learning-based prediction, and performance evaluation.

Railway data, including train identifiers, station information, timestamps, and delay-related features, was organized and preprocessed to support consistent tracking and prediction.

An LSTM-based regression approach was implemented to estimate numerical train delays using sequential data. The research also evaluated a three-class classifier covering On-time / Low Delay, Moderate Delay, and High Delay using precision, recall, F1-score, accuracy, and confusion matrices.

The classifier assigned all 1,065 test observations to Moderate Delay, achieving approximately 34.5% accuracy but failing to identify the other two classes correctly. This demonstrated that accuracy alone is insufficient to establish model suitability and highlighted the need for further refinement.

Overall, the research established the core processing and prediction workflow and evaluated its strengths and limitations.

6.2 Product and Technical Results

The completed system provides a structured workflow for processing railway information, organizing train movement records, tracking train status, predicting delays, and evaluating model performance.

6.2.1 Train-Tracking Results

The tracking component identifies the latest valid observation for a train and presents its status with the corresponding timestamp. Train identifiers, station information, and movement timestamps are maintained to ensure correct record associations.

Recorded movement information is distinguished from predicted arrival information, helping users interpret the latest available status and future estimates. Tracking accuracy and freshness depend on the availability and quality of underlying observations.

6.2.2 Delay-Prediction Results

An LSTM-based regression model was implemented to estimate numerical train delays. Its performance was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). MAE measures the average absolute prediction error, while RMSE gives greater weight to larger errors. When delays are measured in minutes, both metrics can be interpreted in minutes.

The predicted delay can be combined with the scheduled arrival time to calculate the estimated arrival time:

$\hat{T}_a = T_s + \hat{D}$ where,

$\hat{T}_a$: estimated arrival time
T_s : scheduled arrival time
$\hat{D}$: predicted delay

The regression experiment supports the evaluation of numerical delay prediction. However, the exact MAE and RMSE values must be obtained from the original model evaluation output.

6.2.3 Classification Results

The classification model was evaluated on 1,065 observations across three delay categories. Table 6.1 summarizes the reported results.

Table 12 - Classification performance results

Delay class	Precision	Recall	F1-score	Support
On-time / Low Delay	0.000	0.000	0.000	212
Moderate Delay	0.345	1.000	0.513	367
High Delay	0.000	0.000	0.000	486
Macro average	0.115	0.333	0.171	1,065
Weighted average	0.119	0.345	0.177	1,065

The reported accuracy was approximately 34.5%. The confusion matrix showed that all 212 On-time / Low Delay, 367 Moderate Delay, and 486 High Delay observations were predicted as Moderate Delay. Consequently, only the 367 actual Moderate Delay observations were correctly classified, while the remaining 698 were misclassified.

The normalized confusion matrix confirmed that 100% of observations in every actual class were assigned to Moderate Delay. Therefore, the classifier cannot reliably generate differentiated delay alerts in its current state.

This limitation does not invalidate the overall implementation but identifies the need to investigate label preparation, class distribution, feature processing, and model configuration, followed by further testing.

6.2.4 Technical Achievements

The main technical achievements include:

- Developing a structured railway data-loading and preprocessing workflow.
- Organizing train and station information for consistent tracking.
- Implementing tracking functionality using available movement observations.
- Implementing and evaluating an LSTM-based regression experiment.
- Evaluating three-class delay classification using standard metrics.
- Generating confusion matrices to examine actual and predicted classes.
- Identifying classification limitations and areas for improvement.

These achievements demonstrate the implementation of the system's core data-processing, tracking, prediction, and evaluation functions.

6.3 User or Stakeholder Validation

Stakeholder validation considers whether the system meets railway information needs, particularly the clarity of train status, usefulness of estimated arrival times, and reliability of displayed information.

The system separates recorded movement observations from predictions, helping users understand the latest available status and interpret arrival estimates as predictions rather than guarantees.

6.3.1 Validation of Functional Requirements

The system was assessed against its main objectives: preparing railway data, processing movement information, predicting delays, and evaluating model performance.

Table 13 - Functional validation summary

Functional area	Validation outcome
Railway data preparation	Data-processing workflow implemented to organize and prepare relevant railway information
Train and station identification	Identifiers incorporated into the processing workflow to maintain record consistency
Train-status processing	Tracking-related functionality implemented using available movement observations
Delay prediction	LSTM-based regression experiment implemented and evaluated
Delay classification	Three-class classification evaluated; results identified a single-class prediction limitation
Model evaluation	Classification report, confusion matrices, and regression evaluation workflow used to assess performance
Output interpretation	Tracking observations and predicted information treated as distinct types of output

The validation confirms that the principal software functions were implemented and the prediction components quantitatively evaluated. The classification results also demonstrate that producing an output does not necessarily make it useful for its intended purpose.

6.3.2 Stakeholder Relevance and Usability

The system is designed to support passengers and railway-related stakeholders by organizing train movement information and estimating delays. Passengers can benefit from clearer status information and estimated arrival times, while railway stakeholders can examine train movements and potential delays.

Separating processing, tracking, and prediction into distinct stages supports maintenance and targeted improvements. However, the classifier requires further refinement before its delay categories can reliably support operational decisions.

6.3.3 Validation Limitations

Software functionality and predictive accuracy are distinct aspects of validation. Although the tracking, processing, and prediction workflows were implemented, the classification metrics reveal significant predictive limitations.

Tracking usefulness depends on the availability and freshness of observations, which timestamps help users assess. Arrival-time estimates depend on regression

performance, while reliable classification requires further improvement. These limitations must be considered when evaluating the system's practical suitability.

6.4 Integration Results

The integration connects railway data preparation, tracking, prediction, and evaluation into a coordinated workflow, with each stage providing structured inputs to the next.

6.4.1 Data and Processing Integration

The data-processing component standardizes train identifiers, station information, timestamps, and delay-related features before tracking and prediction. Preserving these fields ensures that predictions correspond to the correct train and target station and allows prepared observations to be compared with actual delays or categories during evaluation.

6.4.2 Tracking and Prediction Integration

Tracking organizes the latest available movement observations, while prediction estimates numerical delays or assigns delay categories. Where the required inputs are available, the scheduled arrival time and predicted delay can be combined to calculate an estimated arrival time.

Recorded information remains distinct from model-generated predictions to prevent estimates from being interpreted as confirmed events. However, the classifier's inability to distinguish the three delay categories means its outputs require refinement before supporting differentiated operational alerts.

6.4.3 Integration Reliability

Integration reliability depends on consistent data formats, valid inputs, and correct associations between trains and observations. Missing or invalid timestamps can affect observation ordering, while inconsistent identifiers can associate information with the wrong train. LSTM inputs must also follow the structure used during training.

Separating data preparation, tracking, prediction, and evaluation improves maintainability by allowing individual stages to be investigated and refined independently. The integrated workflow therefore provides a foundation for further development, although connected components must still be evaluated individually to ensure reliable results.

7. DISCUSSION

7.1 Interpretation of Findings

The Real-Time Train Tracking and Delay Prediction system demonstrates the application of data processing and machine learning to railway monitoring. It integrates railway data preparation, train-status processing, delay prediction, and model evaluation into a structured workflow.

The data-processing workflow organizes train identifiers, station information, scheduled and actual times, and delay observations into a consistent structure, supporting subsequent tracking and prediction.

The tracking component identifies the latest available train movement observation and associates it with the relevant train. Combining this information with predicted delays provides a clearer understanding of train status while distinguishing recorded observations from estimated arrival times.

An LSTM-based regression approach was implemented to predict numerical delays by investigating temporal patterns in railway observations. Predicted delays can be combined with scheduled arrival times to estimate arrivals, while regression-error metrics provide a basis for assessing prediction performance.

The three-class classification experiment evaluated On-time / Low Delay, Moderate Delay, and High Delay. However, its accuracy was approximately 34.5%, and all 1,065 test observations were classified as Moderate Delay. Although the classifier failed to distinguish the three categories adequately, the experiment highlighted the importance of class-specific precision, recall, F1-score, and confusion matrices. These results establish a measurable baseline for improvement.

Overall, the project implemented the core tracking and prediction workflow and identified specific opportunities to improve predictive reliability.

7.2 Comparison with Literature Baselines and Existing Solutions

Previous research has investigated railway tracking, arrival-time estimation, and delay prediction using GPS/GSM tracking, historical travel times, online prediction, and machine-learning approaches.

Jayasinghe and Kathriarachchi (2021), in *Real-time Train Tracking System in Sri Lanka*, examined railway tracking in Sri Lanka, addressing the need to make train movement information accessible. Wanniarachchi et al. (2024), in *Real-Time Train*

Tracking and Time Prediction System for Railway System in Sri Lanka, combined tracking and time prediction, closely aligning with the present project's objectives.

Jayakody et al. (2010) investigated GPS/GSM-based train tracking using mobile networks. This research provides a technological background for railway monitoring, while the present project's tracking component focuses on processing available movement observations and associating them with train information.

For delay prediction, Pongnumkul et al. (2014) studied passenger train arrival-time prediction in Thailand using historical travel times. Kecman and Goverde (2013) investigated adaptive, data-driven online prediction of train event times, highlighting the importance of temporal data and updated predictions.

Lapamonpinyo et al. (2022) examined real-time passenger train delay prediction using machine learning in an Amtrak case study. Vafaei and Yaghini (2024) investigated online machine-learning prediction of arrival and departure times at individual stations. More recently, Meesit and Suwannasri (2025) examined real-time delay prediction for single-track intercity railways.

The developed system follows these research directions by integrating data preparation, tracking-related processing, and machine-learning-based delay prediction into a coordinated workflow.

Table 14 - Comparison with related research directions

Research direction	Main focus in the literature	Relationship to the developed system
Real-time train tracking	Monitoring train movements and making movement information available	Supports the tracking and train-status processing objectives
GPS/GSM-based tracking	Acquiring and communicating train-location information	Provides a relevant technological background for railway tracking
Historical travel-time prediction	Using previous travel observations to estimate arrival times	Relates to feature preparation and delay prediction
Online train-event prediction	Updating estimated train-event times using operational data	Relates to the integration of movement information and prediction

Machine-learning-based delay prediction	Estimating train delays using historical and operational features	Provides the research context for the LSTM regression experiment
Developed system	Organizing railway data, processing movement information, implementing delay prediction, and evaluating model performance	Combines the principal software functions within one component

The comparison shows that the project addresses an established research problem through an integrated implementation. However, differences in datasets, operating conditions, and evaluation methods prevent claims of superior performance without comparable experimental results.

The project is therefore best understood as an implementation and evaluation contribution. Its findings, particularly the classifier's limitations, provide a foundation for further experimentation and refinement.

7.3 Research Contribution

The principal research contribution is the implementation and evaluation of a data-driven railway tracking and delay-prediction workflow combining data preparation, tracking, numerical prediction, and classification evaluation.

First, the project establishes a data-processing approach that organizes train identifiers, station information, timestamps, and delay-related features. Consistent input preparation is essential because prediction quality depends on both data relevance and model architecture.

Second, the LSTM-based regression experiment provides a basis for investigating sequential modelling for numerical delay prediction. Its evaluation process compares predicted and actual delays using appropriate regression metrics.

Third, the three-class classification experiment provides evidence of model behavior through classification reports and confusion matrices. Its single-class prediction pattern, in which all test observations were assigned to Moderate Delay, demonstrates why individual class performance must be examined rather than relying solely on accuracy.

Fourth, the integrated workflow connects train movement information with predicted delays, supporting analysis of railway status and estimated arrival times.

Finally, the implementation provides a foundation for investigating additional features, alternative models, classification improvements, and the effects of data quality on predictive performance.

These contributions are primarily practical and experimental, demonstrating established data-processing and machine-learning techniques in a railway application while identifying specific opportunities for improvement.

7.4 Product Contribution

The product contribution is a railway tracking and delay-prediction component that organizes the software functions required to monitor train movements and estimate delays.

The data-preparation component processes train identifiers, timestamps, station information, and delay-related features, providing consistent inputs for tracking and prediction. This structure also improves maintainability.

The tracking functionality organizes available movement observations and associates them with the relevant trains. Timestamps help distinguish recent observations from older records and provide context for interpreting train status.

The LSTM regression experiment adds numerical delay analysis. Predicted delays can be combined with scheduled arrival times to calculate estimated arrivals, potentially providing useful information for passengers and railway stakeholders.

The evaluation component uses classification reports, confusion matrices, and regression metrics to examine predictive outputs, identify limitations, and guide measurable improvements.

Separating data preparation, tracking, prediction, and evaluation allows individual components to be refined without redesigning the entire workflow. For example, the classifier can be improved independently while retaining the existing processing structure.

The component therefore provides both an implemented software workflow and an evaluation foundation for further railway tracking and delay-prediction development. Classification reliability must improve before delay categories are used for operational alerts.

7.5 Limitations and Threats to Validity

Although the core software workflow was implemented, several limitations must be considered when interpreting the results.

7.5.1 Data Availability and Quality

Tracking and prediction reliability depend on data completeness, quality, and consistency. Missing observations, inconsistent timestamps, and incorrectly associated train identifiers can affect preprocessing and model performance.

Although preprocessing organizes available records, generalizability depends on whether the evaluation data adequately represent the railway conditions in which the system may operate.

7.5.2 Classification Performance

The main quantitative limitation was the classifier's tendency to predict Moderate Delay for every test observation. It achieved approximately 34.5% accuracy across 1,065 observations, with zero recall for On-time / Low Delay and High Delay.

Potential areas for investigation include class-label preparation, training-data distribution, feature selection, model configuration, and prediction-label mapping.

This limitation identifies a specific technical problem rather than invalidating the overall implementation. Further refinement and repeated evaluation using a consistent methodology are required to determine whether performance improves across all three classes.

7.5.3 Regression Evaluation

The LSTM regression experiment must be evaluated using numerical error metrics rather than classification accuracy. Its usefulness depends on how closely predicted delays match actual values.

Comparison with a suitable baseline is necessary to determine whether the model improves upon simpler methods. Because railway observations are sequential, chronological train-test splitting can provide a more realistic assessment of predictions for future observations.

7.5.4 Generalizability

Railway operations vary across routes, stations, timetables, traffic conditions, and disruptions. A model evaluated on one dataset may perform differently on another network or during another operating period.

Testing additional routes, time periods, and representative observations would help establish whether the system's processing and prediction methods generalize to other railway environments.

7.5.5 Real-Time Operational Conditions

Practical performance depends on input-data freshness and processing time. Even a rapidly generated prediction may be of limited value if its underlying movement information is outdated.

Further operational assessment should measure data freshness, processing latency, and prediction-generation time under different workloads. Claims concerning continuous GPS tracking or live railway integration must also reflect the actual data sources and interfaces implemented and tested.

7.5.6 Evaluation and Interpretation

Evaluation findings depend on target-variable definitions, selected metrics, and test-data preparation. Incorrect label mappings or inconsistent preprocessing between training and evaluation may affect the interpretation of results.

Using multiple metrics provides a more reliable assessment than relying on a single measure. Classification reports and confusion matrices reveal class-specific behavior, while regression-error metrics assess numerical prediction quality.

These limitations provide context for interpreting the findings and guide further technical development.

7.6 Implications

The project has implications for railway information management, passenger information, predictive modelling, software engineering, and future railway applications.

7.6.1 Implications for Railway Information Management

The project demonstrates the importance of consistently organizing train identifiers, station information, timestamps, and delay-related features. Structured processing supports correct associations between movement records and predictions, helps identify inconsistent information, and provides a foundation for incorporating additional data sources.

7.6.2 Implications for Passenger Information

Clear train-status information and estimated arrival times can help passengers understand expected journey progress. Separating recorded observations from predictions supports clearer interpretation.

However, passenger-facing information depends on data quality and prediction accuracy. The classifier's single-class prediction behavior highlights the need to improve reliability before generating differentiated passenger alerts.

7.6.3 Implications for Predictive Modelling

The project demonstrates that regression and classification require different evaluation methods. The LSTM regression experiment investigates numerical delay prediction, while classification metrics reveal whether individual delay categories are distinguished effectively.

Future development should prioritize reliable feature preparation, suitable training configurations, baseline comparisons, and evaluation across all relevant categories.

7.6.4 Implications for Software Engineering

Separating data processing, tracking, prediction, and evaluation supports testing, maintenance, and extension. Integrating data validation with model evaluation also emphasizes that successful execution alone is insufficient: outputs must be reliable and appropriately interpreted.

This structure supports further improvements in error handling, model version management, and operational testing.

7.6.5 Implications for Future Railway Applications

The developed component provides a foundation for combining movement monitoring with predictive analysis. Future development could incorporate additional operational observations, refine classification, compare alternative regression approaches, and represent uncertainty in arrival-time estimates.

8. CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research developed a Real-Time Train Tracking and Delay Prediction component combining railway data processing, train tracking, machine-learning-based prediction, and performance evaluation within a structured workflow.

The data-processing workflow organizes train identifiers, station information, timestamps, and delay-related features, providing a consistent foundation for tracking and prediction.

The tracking component identifies the latest available train movement observation and distinguishes recorded information from predicted values. Its reliability depends on the availability and freshness of the underlying data.

An LSTM-based regression experiment was implemented to estimate numerical train delays using appropriate regression metrics. Predicted delays can be combined with scheduled arrival times to calculate estimated arrival times when the required information is available.

The three-class classification experiment evaluated On-time / Low Delay, Moderate Delay, and High Delay using precision, recall, F1-score, accuracy, and confusion matrices. However, all 1,065 test observations were classified as Moderate Delay, identifying a significant limitation in distinguishing delay conditions.

Despite this limitation, the project achieved its main implementation objective by establishing a coordinated workflow for data preparation, tracking, prediction, and evaluation. The results provide a foundation for improving prediction reliability, data quality, and operational suitability.

8.2 Achievement of Objectives

The project addressed seven objectives covering data preparation, processing, tracking, prediction, evaluation, integration testing, and future improvements.

Objective 1: To identify and prepare the railway data required for tracking and delay prediction, including available timetable, timestamp, route, station, and delay information.

Railway identifiers, timestamps, station information, and delay-related features were organized to support consistent processing and establish the foundation for tracking and prediction.

Objective 2: To design a data-processing workflow that validates and aligns train identifiers, location observations, timestamps, and relevant operational features.

A structured workflow was implemented to maintain consistency across railway records, associate movement observations with the correct trains, and prepare prediction inputs.

Objective 3: To implement a train-tracking function that presents the most recent available train position or movement status and indicates the data source and freshness where possible.

The tracking functionality identifies the latest available movement observation associated with a train. Timestamps help communicate observation freshness, although reliability depends on the quality and availability of the underlying data.

Objective 4: To develop and evaluate suitable delay-prediction models using selected approaches, such as Linear Regression, Random Forest Regression, and/or LSTM, where these have actually been trained.

An LSTM-based regression experiment was implemented for numerical delay prediction, alongside a separate three-class delay-classification experiment. The final thesis should identify only models actually trained and evaluated.

Objective 5: To assess model performance using appropriate evaluation metrics.

Regression evaluation used Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), where available. Classification evaluation used precision, recall, F1-score, accuracy, and confusion matrices. The classifier assigned all 1,065 test observations to Moderate Delay, identifying a specific limitation and an opportunity for improvement.

Objective 6: To test the tracking and prediction workflow, document its integration, and identify limitations related to data availability, model performance, latency, and deployment.

Data preparation, tracking, prediction, and evaluation were organized into connected stages. The implementation highlighted the importance of consistent inputs, prediction reliability, and observation freshness, providing a basis for identifying technical limitations.

Objective 7: To propose realistic improvements that could strengthen reliability, uncertainty communication, and suitability for future use in railway operations.

Proposed improvements include refining classification, strengthening data validation, expanding model evaluation, incorporating additional operational data, testing across routes and periods, and communicating uncertainty in estimated arrival times.

Overall, the project addressed its principal objectives through implementation and evaluation of the core workflow, while identifying further work needed to improve predictive performance and operational reliability.

8.3 Individual Research and Product Contributions

The individual contribution comprises the design, implementation, and evaluation of a railway tracking and delay-prediction component integrating data preparation, movement-status processing, predictive modelling, and performance evaluation.

8.3.1 Individual Research Contribution

The research applied data-processing and machine-learning techniques to railway tracking and delay prediction through a structured workflow.

The LSTM-based regression experiment investigated numerical delay estimation, while the separate classification experiment examined delay-category prediction. Task-appropriate evaluation metrics provided insight into model behaviour and prediction challenges.

The classifier's single-class prediction pattern demonstrated the importance of assessing individual class performance rather than relying solely on overall accuracy. This finding provides a specific basis for further experimentation and refinement.

The research therefore contributes an implementation-oriented investigation supported by systematic evaluation and documentation of the component's strengths and limitations.

8.3.2 Individual Product Contribution

The product contribution is a software component supporting railway data preparation, train tracking, delay prediction, and model evaluation.

Its processing workflow organizes railway information, the tracking functionality handles available movement observations, and the prediction component estimates numerical delays. Evaluation functions assess predictive quality and identify areas for improvement.

The structured design allows individual stages to be refined while preserving the overall workflow. The component provides a foundation for making railway movement information more accessible and supporting delay estimation.

8.3.3 Overall Contribution

The project establishes an implemented and evaluated workflow connecting railway data processing, train tracking, and delay prediction. The research evaluates predictive approaches, while the product provides the software structure for processing railway information and assessing predictions.

Together, these contributions establish a foundation for further development of data-driven railway monitoring systems.

8.4 Future Directions

Although the core tracking and prediction workflow was implemented, further work is needed to improve accuracy, reliability, and practical suitability.

8.4.1 Improving Prediction Model Performance

The classifier's tendency to assign every test observation to Moderate Delay requires investigation of class labels, class distribution, feature selection, training procedures, and model configuration.

Alternative classifiers, including Random Forest and gradient-boosting methods, could be compared using consistent data splits and evaluation procedures.

For numerical prediction, the LSTM model could be compared with historical average delay, Linear Regression, and tree-based regression methods to determine whether sequential modelling provides measurable benefits.

8.4.2 Expanding and Improving Railway Data

Prediction reliability could improve through additional data, including historical inter-station travel times, scheduled dwell times, previous delays, route characteristics, and available operational conditions.

Larger, representative datasets could support comparisons across routes, stations, peak and off-peak periods, and disruption types.

Data validation should systematically detect missing values, inconsistent timestamps, duplicate records, and incorrect train or station associations. Consistent preprocessing during training and prediction is also essential.

8.4.3 Strengthening Real-Time Tracking

Future development could support more frequent updates from live railway sources where authorized interfaces and suitable infrastructure are available.

Observation-freshness indicators, data-source information, and last-updated timestamps could help users distinguish recent records from older information. Testing different update intervals would help measure tracking latency under realistic conditions.

8.4.4 Improving Estimated Arrival Times

Arrival-time estimation could be improved by incorporating station-specific travel patterns and updated movement observations.

Validated uncertainty ranges or confidence information could communicate the reliability of estimates more clearly. Predictions could also be updated as new observations become available, allowing arrival estimates to respond to changing train movements.

8.4.5 Expanding Evaluation and Validation

Further evaluation should cover different routes, stations, and periods. Chronological train-test splits would help assess performance on observations occurring after training.

Regression experiments should report MAE and RMSE alongside suitable baselines. Classification evaluation should include per-class precision, recall, F1-score, confusion matrices, and accuracy, with particular attention to correctly identifying all three delay categories.

Additional testing should examine processing errors, missing inputs, invalid timestamps, latency, and behavior when observations are unavailable. Where stakeholder feedback or acceptance testing is conducted, its findings should also be documented.

8.4.6 Enhancing System Integration and Deployment

The component could integrate additional railway information sources and passenger-facing interfaces, subject to data availability, authorization, and technical feasibility.

Potential enhancements include automated updates, improved error reporting, model-version management, and continuous performance monitoring.

Live deployment would require broader operational testing, security controls, reliable data interfaces, and validation against operational requirements. Predictions should support decisions rather than replace authoritative railway information.

8.4.7 Long-Term Development

The component could evolve into a broader railway decision-support system by combining reliable movement observations, historical operational data, and validated prediction models.

Future research could explore adaptive prediction, uncertainty-aware forecasting, and methods for identifying how delays develop across successive stations.

The long-term objective is to develop a thoroughly validated system that provides reliable, understandable, and timely information to railway users and stakeholders.

In conclusion, the project establishes a practical foundation for integrating train tracking and delay prediction. Improving model performance, expanding data, strengthening real-time processing, and conducting broader validation could increase its reliability and practical value for railway monitoring and passenger information services.

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GLOSSARY

Term	Definition
Accuracy	The proportion of predictions that are correct out of all predictions made.
Actual Arrival Time	The recorded time at which a train arrives at a station.
Artificial Intelligence (AI)	Techniques that enable computer systems to perform tasks such as prediction and decision-making.
Classification	A machine learning task that assigns data to predefined categories, such as On-time, Moderate Delay, or High Delay.
Confusion Matrix	A table used to compare actual and predicted classes to evaluate a classification model.
Data Cleaning	The process of identifying and correcting missing, inconsistent, duplicate, or invalid data.
Data Preprocessing	Preparing raw data for analysis or machine learning by cleaning, transforming, and organizing it.
Delay Prediction	The process of estimating how late a train may arrive or depart compared with its scheduled time.
Estimated Time of Arrival (ETA)	The predicted time at which a train is expected to arrive at a particular station or destination.
Feature	An input variable used by a machine learning model to generate a prediction.
F1-Score	A classification evaluation metric that combines precision and recall into a single score.
Global Positioning System (GPS)	A satellite-based system used to determine geographic location and support train tracking.
Historical Data	Previously recorded information, such as train schedules, arrival times, departure times, and delays.
Integration	The process of connecting the train-tracking and delay-prediction component with other parts of the overall system.
Long Short-Term Memory (LSTM)	A type of recurrent neural network designed to learn patterns in sequential data, such as changes in train movement over time.
Machine Learning (ML)	A branch of artificial intelligence in which models learn patterns from data to make predictions or classifications.
Mean Absolute Error (MAE)	A regression metric that calculates the average absolute difference between predicted and actual values.
Model Evaluation	The process of measuring how effectively a trained model performs using suitable metrics and test data.

Precision	The proportion of correctly predicted positive instances among all instances predicted as positive for a class.
Prediction	An estimated outcome generated by a model using available data.
Recall	The proportion of actual instances of a class that the model correctly identifies.
Regression	A machine learning task used to predict numerical values, such as arrival times or delay durations.
Real-Time Tracking	Monitoring a train's latest available location or movement status as new information becomes available.
Root Mean Squared Error (RMSE)	A regression metric that calculates the square root of the average squared difference between predicted and actual values.
Scheduled Arrival Time	The planned time at which a train is expected to arrive at a station according to the timetable.
Timestamp	A date and time associated with a recorded event or observation.
Timetable	The planned schedule of train services, including arrival and departure times at stations.
Train Identifier	A unique or designated reference used to distinguish a train or train service within the system.
Training Data	Data used to teach a machine learning model to identify patterns and relationships.
Testing Data	Data reserved for evaluating a model's performance on observations not used for training.
Validation	The process of checking whether a system or model meets its intended requirements and performs as expected.
Data Freshness	An indication of how recently the displayed train location or operational information was updated.
Delay Category	A predefined group representing the severity of a train delay, such as On-time / Low Delay, Moderate Delay, or High Delay.

APPENDIX

Appendix A : UI interfaces

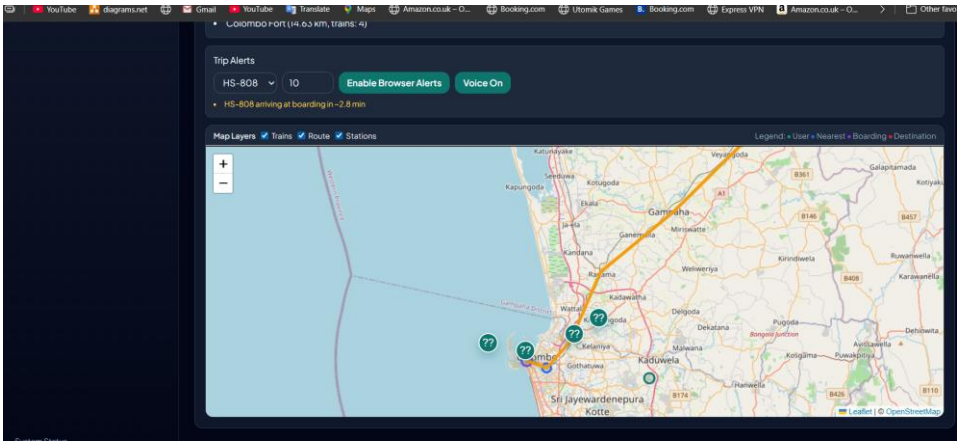


Figure 13 - User Interface 1

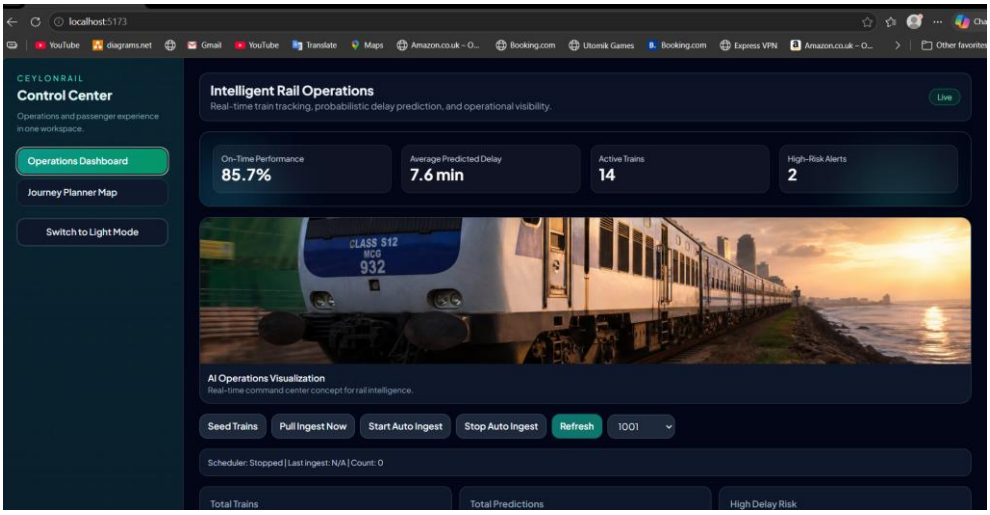


Figure 14 -User Interface 2

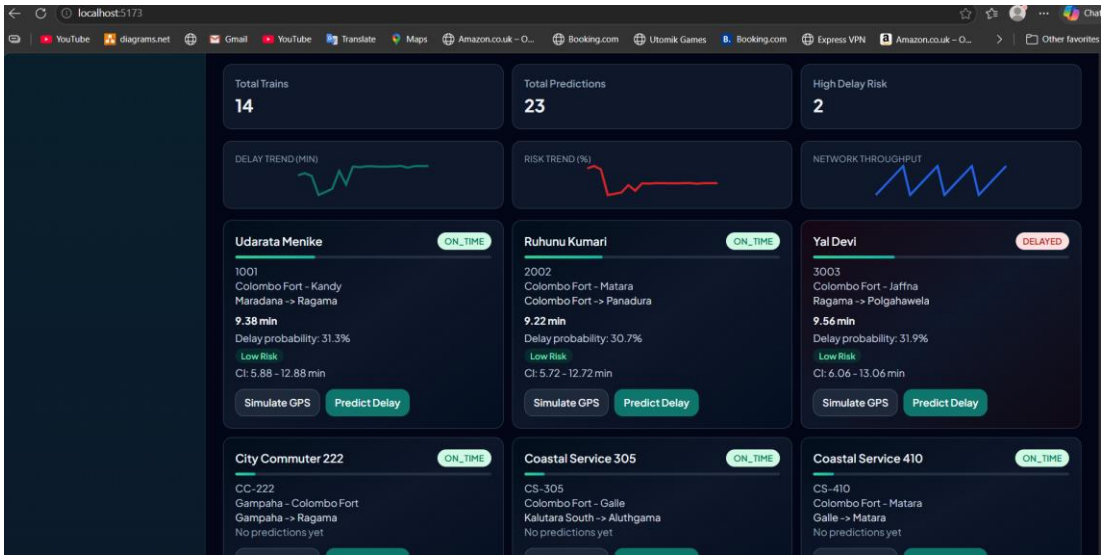
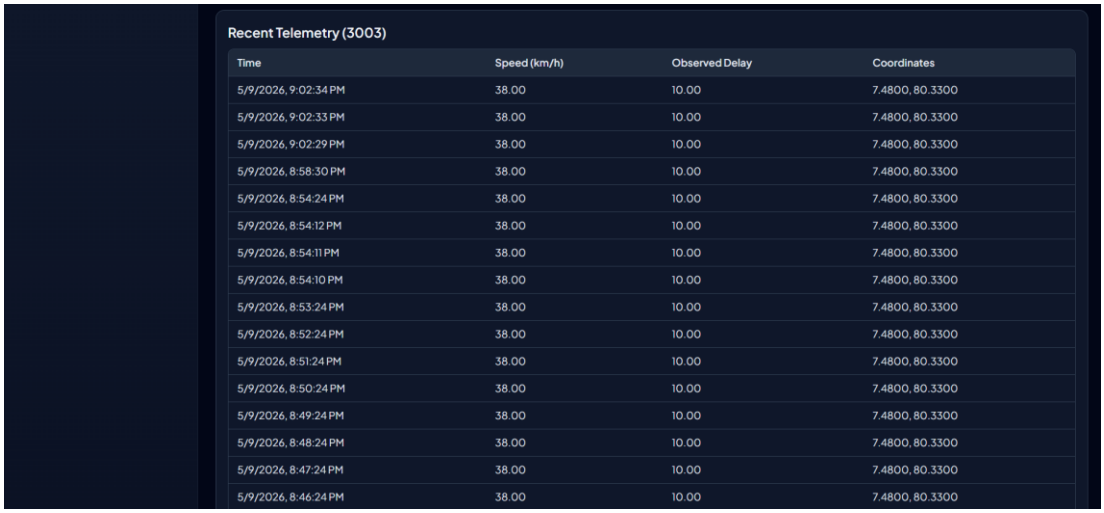


Figure 15 -User Interface 3



Recent Telemetry (3003)

Time	Speed (km/h)	Observed Delay	Coordinates
5/9/2026, 9:02:34 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 9:02:33 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 9:02:29 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:58:30 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:12 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:11 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:10 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:53:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:52:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:51:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:50:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:49:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:48:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:47:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:46:24 PM	38.00	10.00	7.4800, 80.3300

Figure 16 - User Interface 4

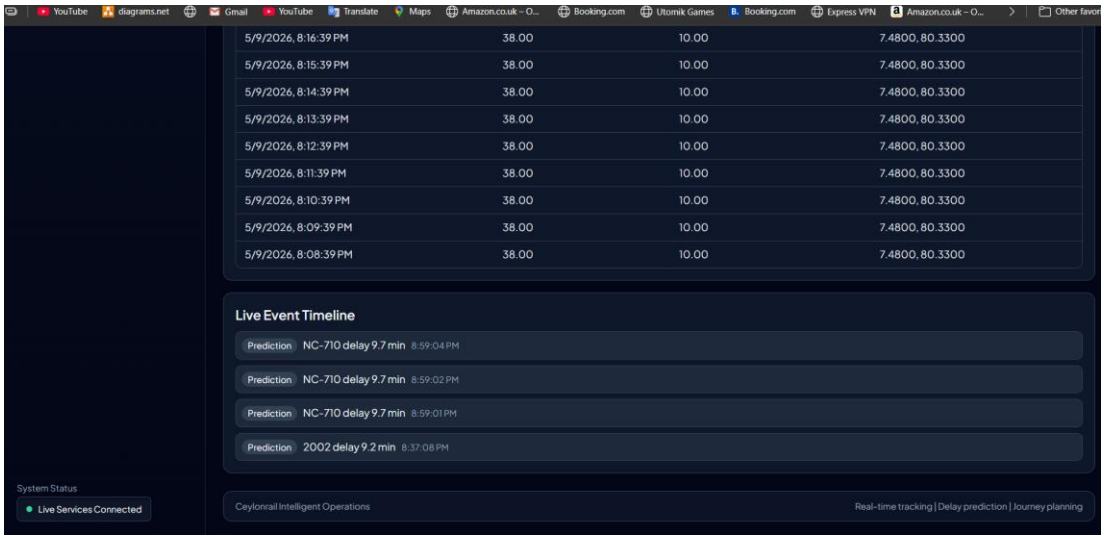


Figure 17 - User Interface 5

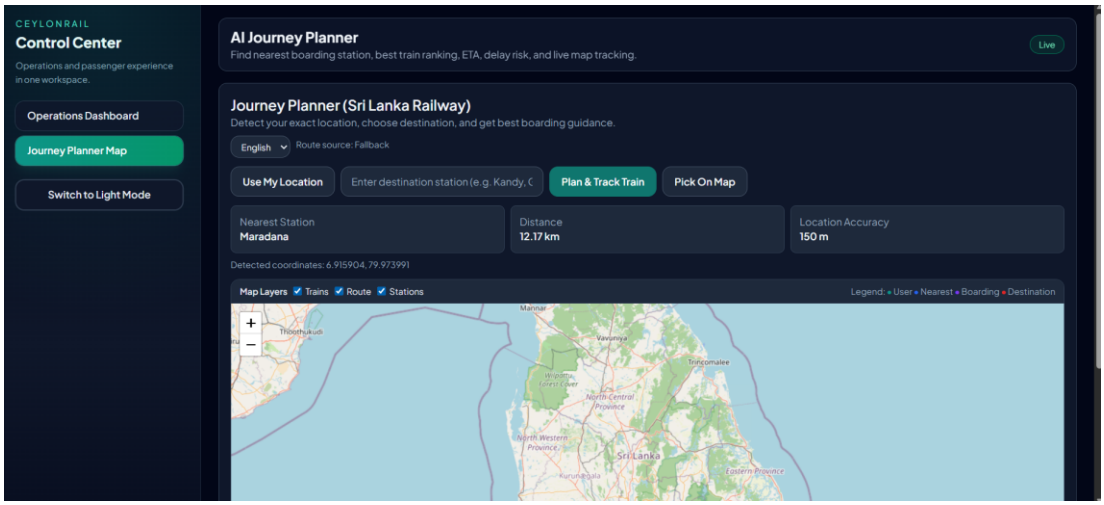


Figure 18 - User Interface 6

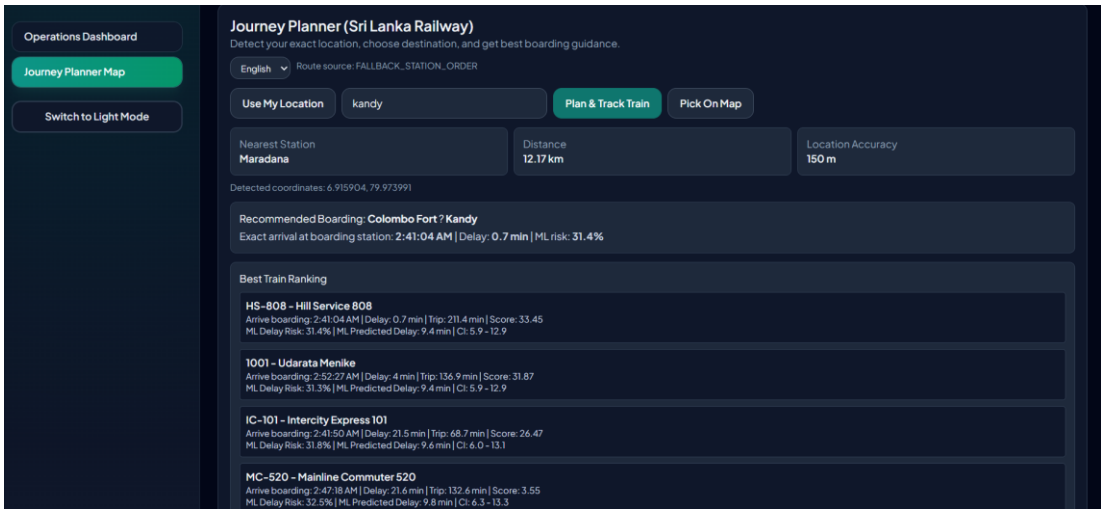


Figure 19 - User Interface 7